# Block practice · Core investigation skills

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Identify the independent or dependent variable

State or explain safety precautions for a practical

Improve reliability with repeats, means and larger samples

State the variables to control in an investigation

Describe a control or explain why a step is in a method

Identify a measurement weakness and improve accuracy

Plan or specify a single-factor laboratory method

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# Practice workbook · Identify the independent or dependent variable

A described method or results table where the student names the factor deliberately changed, the quantity measured, or both.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-PRAC-IDENTIFY-INDEPENDENT-DEPENDENT-VARIABLE

# 1. Question 4 · part b.ii · 2 marks

June 2022 Paper 1R

Worked-example retry · Attempt part b.ii only.

**4** (a) Lipids are digested in the small intestine.

(i) Describe a test for lipids in food.

**(3)**

(ii) Name the enzyme that digests lipids.

**(1)**

(iii) Name the products of lipid digestion.

**(2)**

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(b) A student investigates how the pH of a lipid solution, containing an enzyme that digests lipids, changes over 10 minutes at two different temperatures.

The tables show the student's results.

**Temperature at 37 °C**

| Time in minutes | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| pH | 8.5 | 8.0 | 7.5 | 7.2 | 7.0 | 6.8 | 6.2 | 5.8 | 5.5 | 5.5 |

**Temperature at 80 °C**

| Time in minutes | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| pH | 9.2 | 9.0 | 8.8 | 8.8 | | 8.6 | 8.6 | 8.4 | 8.4 | 8.4 |

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# 1. Question 4 · part b.ii · 2 marks

June 2022 Paper 1R

Worked-example retry · Attempt part b.ii only.

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(i) Estimate a pH value for the lipid solution at 80 °C at 5 minutes.

**(1)**

(ii) Give the independent variable and the dependent variable for this investigation.

**(2)**

independent

dependent

(iii) Explain the differences in the results of the student's investigation at the two different temperatures.

**(4)**

**(Total for Question 4 = 13 marks)**

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## 1. Question 4 · part b.ii · 2 marks · Working and answer

Working and answer

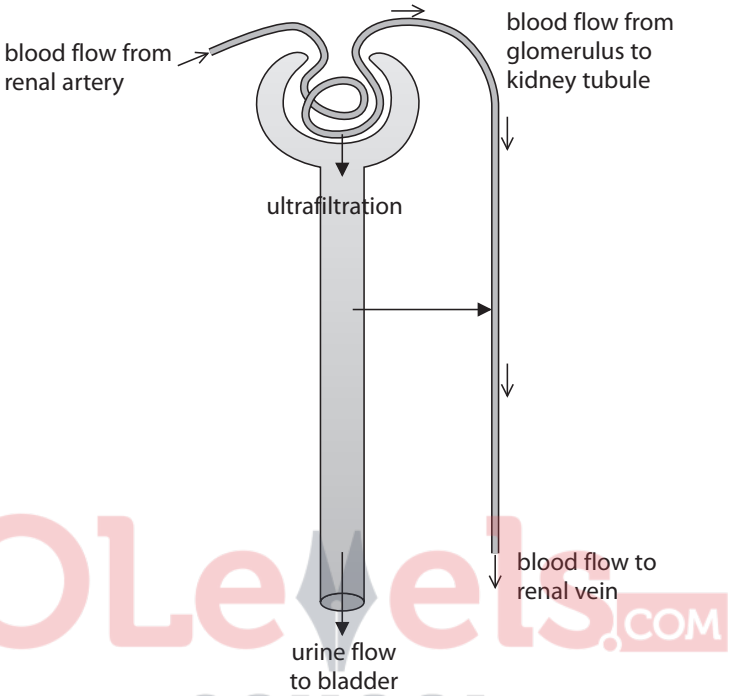
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# 2. Question 8 · part b.i · 1 mark

June 2019 Paper 2

Further practice · Attempt part b.i only.

**8** (a) The diagram shows part of a nephron in the human excretory system.




(Source: adapted from © Alila Medical Images/Alamy)

(i) Explain how ultrafiltration occurs in the Bowman's capsule.

**(2)**

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**20**

## 2. Question 8 · part b.i · 1 mark

June 2019 Paper 2

Further practice · Attempt part b.i only.

(ii) Explain the differences in the composition of the blood flowing through the renal artery and through the renal vein.

**(4)**

(b) The quantity of salt in the diet affects the concentration of urine produced.

An investigation is set up to test this statement.

This is the method used.

- ten people are each given a different food
- each food contains a different quantity of salt
- the volume of urine produced by each person is recorded each hour for six hours after eating the food

(i) State the independent variable in this investigation.

**(1)**

(ii) Explain why this investigation is unlikely to produce valid results.

**(2)**

(iii) State one change to the method that would make the results more valid.

**(1)**

**(Total for Question 8 = 10 marks)**

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## 2. Question 8 · part b.i · 1 mark · Working and answer

Working and answer

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# 3. Question 4 · part a.iii · 1 mark

November 2020 Paper 2

Further practice · Attempt part a.iii only.

**4** Aspirin is the active ingredient used in some tablets to relieve pain.

Aspirin becomes acidic when it dissolves in the body. The faster aspirin dissolves, the faster it can relieve pain.

A student investigates if it is better to mix sucrose or starch with the active ingredient to make an aspirin tablet.

This is the student's method.

- add distilled water to three beakers, A, B and C
- add 1 aspirin tablet to each beaker
- add 5 g of sucrose to beaker B
- add 5 g of starch to beaker C
- measure the pH of the contents of each beaker at the start of the investigation
- measure the pH of the contents of each beaker every minute for 5 minutes

The table shows the student's results.

| Beaker | pH of contents of beaker | | | | | |
|---|---|---|---|---|---|---|
| | at start | 1 minute | 2 minutes | 3 minutes | 4 minutes | 5 minutes |
| A – aspirin | 7.6 | 7.1 | 6.6 | 6.2 | 5.8 | 5.4 |
| B – aspirin and sucrose | 7.2 | 6.7 | 6.3 | 5.7 | 5.2 | 4.7 |
| C – aspirin and starch | 7.1 | 7.2 | 7.3 | 7.3 | 7.4 | 7.4 |

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## 3. Question 4 · part a.iii · 1 mark

November 2020 Paper 2

Further practice · Attempt part a.iii only.

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(a) (i) Plot a graph of the student's results.

Join the points with straight lines.

**(5)**



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(ii) State two variables that should be controlled in this investigation.

**(2)**

1 .....

2 .....

(iii) Name the dependent variable in this investigation.

**(1)**

.....

## 3. Question 4 · part a.iii · 1 mark

November 2020 Paper 2

Further practice · Attempt part a.iii only.

(b) Explain how the results of this investigation could help a manufacturer improve the effectiveness of its aspirin tablets.

**(3)**

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**(Total for Question 4 = 11 marks)**

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Working and answer

# 4. Question 7 · part a.ii · 1 mark

June 2021 Paper 2

Further practice · Attempt part a.ii only.

**7** Garlic plants contain a substance called allicin. This substance is thought to have antibacterial properties.

A student investigates the antibacterial effect of garlic.

This is the student's method.

- label three test tubes A, B and C
- add 10 $cm^3$ of milk into each of the tubes
- add a sample of the bacterium *E. coli* into tubes B and C using aseptic techniques
- crush the garlic in water to form garlic juice
- add 1 $cm^3$ of garlic juice to tube C
- shake all three tubes and leave to incubate for two hours
- place a sample from tube A into the centre of Petri dish A, from tube B into the centre of Petri dish B and from tube C into the centre of Petri dish C
- incubate the three Petri dishes for five days
- each day measure the diameter of any bacterial growth that has occurred

The table shows the student's results.

| Petri dish | Diameter of bacterial growth in mm | | | | |
|---|---|---|---|---|---|
| | day 1 | day 2 | day 3 | day 4 | day 5 |
| A | 0 | 1.7 | 3.0 | 4.6 | 7.1 |
| B | 0 | 4.2 | 8.4 | 15.1 | 36.5 |
| C | 0 | 0 | 0 | 0 | 0 |

(a) (i) State three control variables in the student's investigation.

**(3)**

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# 4. Question 7 · part a.ii · 1 mark

June 2021 Paper 2

Further practice · Attempt part a.ii only.

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(ii) State the dependent variable in the student's investigation.

**(1)**

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(b) (i) Calculate the mean daily increase in the diameter of the bacterial growth for dish A and dish B.

**(2)**

mean daily increase for dish A ..... mm/day

mean daily increase for dish B ..... mm/day

(ii) Explain the difference between the mean daily increases in the diameter of the bacterial growth for dish A and dish B.

**(4)**

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(c) Explain what is meant by an aseptic techniques for the transfer of bacteria.

**(2)**

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## 4. Question 7 · part a.ii · 1 mark · Working and answer

Working and answer

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# 5. Question 6 · part a · 1 mark

June 2022 Paper 2

Further practice · Attempt part a only.

**6** A student uses this method to investigate the effect of different antibacterial cleaners on the growth of bacteria.

- add bacterial culture to sterile agar in a Petri dish using an inoculating loop
- using a pipette, place two drops of an antibacterial cleaner onto a filter paper disc
- repeat using two other filter paper discs, each one soaked with a different antibacterial cleaner
- soak one filter paper disc in distilled water
- place the four filter paper discs onto the sterile agar
- place a lid on the Petri dish
- place the dish into an incubator at 25 °C for 5 days

(a) State the independent variable in this investigation.

**(1)**

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(b) Give two improvements to this method that will reduce the risk of introducing unwanted microorganisms into the Petri dish.

**(2)**

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## 5. Question 6 · part a · 1 mark · Working and answer

Working and answer

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## 6. Question 5 · part a.ii · 1 mark

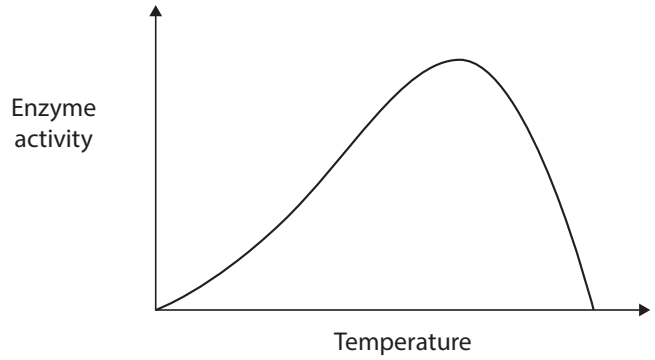
June 2024 Paper 1

Further practice · Attempt part a.ii only.

**5** (a) A group of students investigate the effect of temperature on enzyme activity.

The students use the results they obtain to plot a graph.

The graph shows the students' results.




(Source: adapted from https://ib.bioninja.com.au/standard-level/topic-2-molecular-biology/25-enzymes/enzyme-activity.html)

(i) Design a method the students could use to obtain the results shown in the graph.

**(5)**

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## 6. Question 5 · part a.ii · 1 mark

June 2024 Paper 1

Further practice · Attempt part a.ii only.

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(ii) State the independent variable in this investigation.

**(1)**

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(b) Explain how enzyme activity is affected by low temperatures.

**(4)**

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**(Total for Question 5 = 10 marks)**

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Working and answer

## 7. Question 2 · part c · 1 mark

November 2024 Paper 1

Further practice · Attempt part c only.

**2** A student investigates the activity of an enzyme on fat found in milk.
This is the student's method.

- pour milk and enzyme solution into a test tube
- determine the pH of the solution every minute for 10 minutes

The table shows the student's results.

| Time in minutes | pH of solution |
|---|---|
| 1 | 6.8 |
| 2 | 6.4 |
| 3 | 6.0 |
| 4 | 5.8 |
| 5 | 5.6 |
| 6 | 5.2 |
| 7 | 5.0 |
| 8 | 4.8 |
| 9 | 4.6 |
| 10 | 4.6 |

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# 7. Question 2 · part c · 1 mark

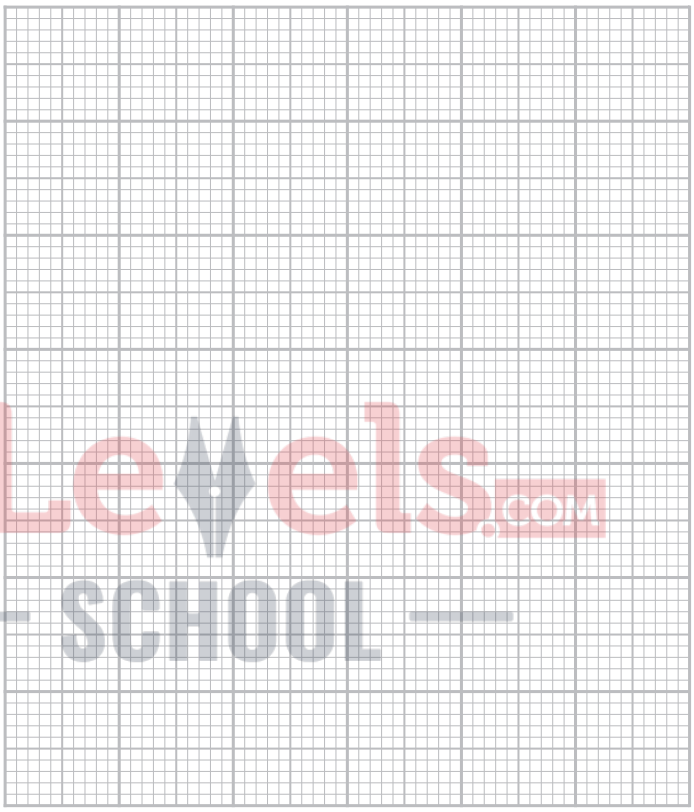
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Further practice · Attempt part c only.

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(a) (i) Plot a graph of the student's results.

**(3)**



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(ii) Draw a line of best fit.

**(1)**

(iii) Give the trend shown by the graph.

**(1)**

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## 7. Question 2 · part c · 1 mark

November 2024 Paper 1

Further practice · Attempt part c only.

(b) Explain the difference in the pH at 1 minute and the pH at 10 minutes.

**(3)**

(c) Name the independent variable in this investigation.

**(1)**

(d) Give two variables that the student should control in this investigation.

**(2)**

1 .....

2 .....

(e) State how the student could improve the reliability of their results.

**(1)**

**(Total for Question 2 = 12 marks)**

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## 7. Question 2 · part c · 1 mark · Working and answer

Working and answer

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## 8. Question 3 · part b.ii · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.ii only.

**3** (a) The nervous system is made up of sensory, motor and relay neurones.

(i) Draw a diagram of a motor neurone.

You should label these structures in your diagram.

- nucleus
- axon
- myelin sheath

**(4)**

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(ii) Describe the function of a motor neurone.

**(2)**

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# 8. Question 3 · part b.ii · 1 mark

June 2025 Paper 2R

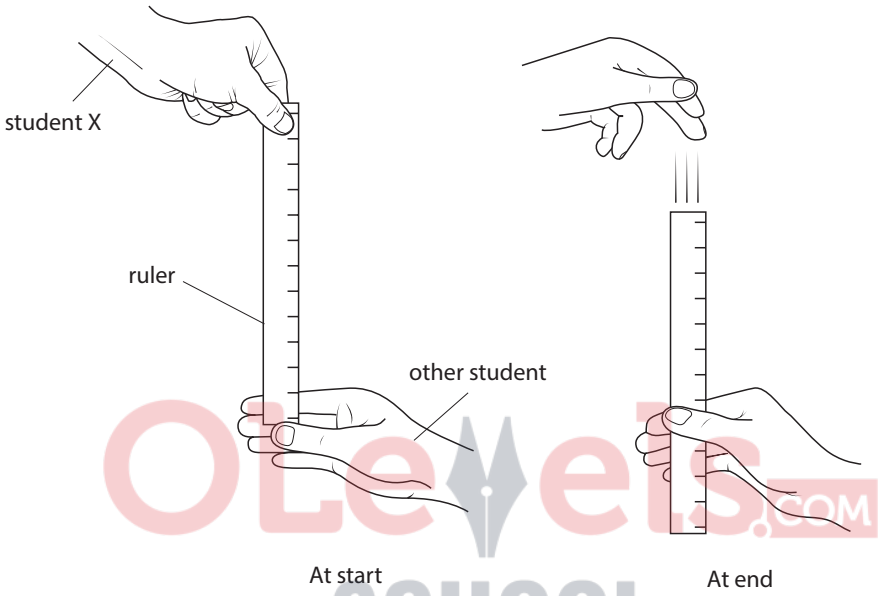
Further practice · Attempt part b.ii only.

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(b) Student X uses this equipment to investigate the reaction time of another student.




Student X knows that the reaction time can be calculated using this formula.

$$\text{reaction time} = \sqrt{\frac{\text{measurement from ruler}}{5}}$$

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# 8. Question 3 · part b.ii · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.ii only.

(i) Describe a method the student could use to compare the reaction times of 10 students.

**(6)**

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(ii) Name the dependent variable in this investigation.

**(1)**

(iii) Give a reason why the student should display the results using a bar chart instead of a line graph.

**(1)**

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## 8. Question 3 · part b.ii · 1 mark · Working and answer

Working and answer

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## 9. Question 7 · part b · 2 marks

November 2024 Paper 2

Further practice · Attempt part b only.

**7** Six students investigate the activity of an enzyme found in the alimentary canal.

This enzyme digests a cloudy suspension into a clear solution.

The results of the investigation are shown in the table.

| Student | Volume of enzyme solution in $cm^3$ | Temperature in °C | pH | Volume of cloudy suspension in $cm^3$ | Time tubes left in minutes | Appearance at end of investigation |
|---|---|---|---|---|---|---|
| 1 | 5 | 37 | 2 | 10 | 9 | cloudy |
| 2 | 5 | 37 | 2 | 10 | 20 | clear |
| 3 | 5 | 37 | 4 | 10 | 35 | clear |
| 4 | 5 | 37 | 4 | 10 | 20 | cloudy |
| 5 | 5 | 37 | 6 | 10 | 8 | cloudy |
| 6 | 5 | 37 | 6 | 10 | 40 | cloudy |

(a) Explain why the investigation was carried out at 37 °C.

**(2)**

(b) Explain which property of the enzyme the students are investigating.

**(2)**

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## 9. Question 7 · part b · 2 marks

November 2024 Paper 2

Further practice · Attempt part b only.

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(c) Comment on the results of the investigation.

In your answer include ideas of validity and accuracy.

**(4)**

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(d) Describe a control experiment that the students could have carried out.

**(3)**

## 9. Question 7 · part b · 2 marks · Working and answer

Working and answer

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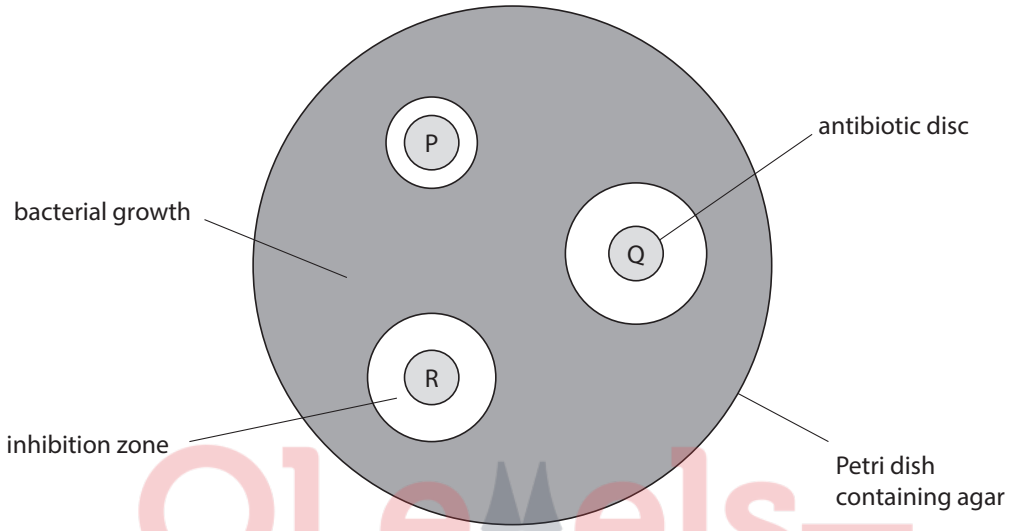
## 10. Question 5 · part a.i · 1 mark

June 2024 Paper 2

Further practice · Attempt part a.i only.

**5** (a) A student investigates the action of three different antibiotics, P, Q and R, on the growth of bacteria.

The diagram shows the results of the investigation.




(i) Which is the independent variable in this investigation?

**(1)**

- ☒ **A** the mass of bacteria added
- ☒ **B** the size of the inhibition zone
- ☒ **C** the type of antibiotic added
- ☒ **D** the mass of agar in the Petri dish

(ii) Give two procedures that the student should use to reduce the risk of contamination by unwanted bacteria.

**(2)**

1 .....

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2 .....

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P 7 4 6 3 0 A 0 1 4 2 4

## 10. Question 5 · part a.i · 1 mark · Working and answer

Working and answer

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# Practice workbook · State or explain safety precautions for a practical

Burning food, hot or cold water, reagents, glass, mouthpieces, spirometers and handling cells or bacteria: hazards and how to reduce risk.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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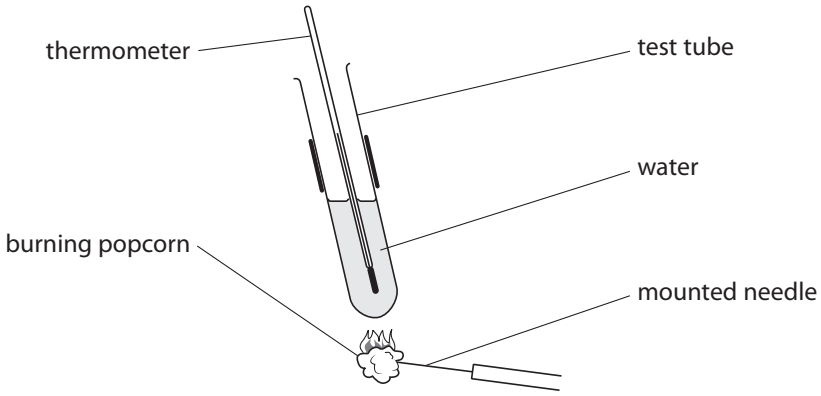
Type ID: HBI-PRAC-SAFETY-HAZARD-PRECAUTION

# 1. Question 2 · part a.ii · 2 marks

November 2020 Paper 1

Worked-example retry · Attempt part a.ii only.

**2** A student uses this apparatus to investigate the energy contained in popcorn.




This is the student's method.

- place 12 g of water in a test tube
- measure the initial temperature of the water
- place the popcorn on a mounted needle
- light the popcorn using a Bunsen flame
- place the burning popcorn under the test tube
- measure the maximum temperature of the water

(a) (i) State why the test tube should have thin glass walls.

**(1)**

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(ii) Explain one safety precaution that the student should take during the investigation.

**(2)**

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# 1. Question 2 · part a.ii · 2 marks

November 2020 Paper 1

Worked-example retry · Attempt part a.ii only.

(b) The student uses this equation to calculate the energy released by the popcorn.

energy released = mass (g) × 4.2 × temperature rise (°C)

The initial temperature of the 12 g of water is 17 °C.

The maximum temperature of the water is 45 °C.

(i) Calculate the energy released when the student burns the popcorn.

Give your answer to 2 significant figures.

**(3)**

energy released = ..... J

(ii) Explain why the calculated amount of energy released is not the same as the full amount of energy contained in the popcorn.

**(3)**

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## 1. Question 2 · part a.ii · 2 marks · Working and answer

Working and answer

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# 2. Question 3 · part d · 2 marks

June 2019 Paper 2

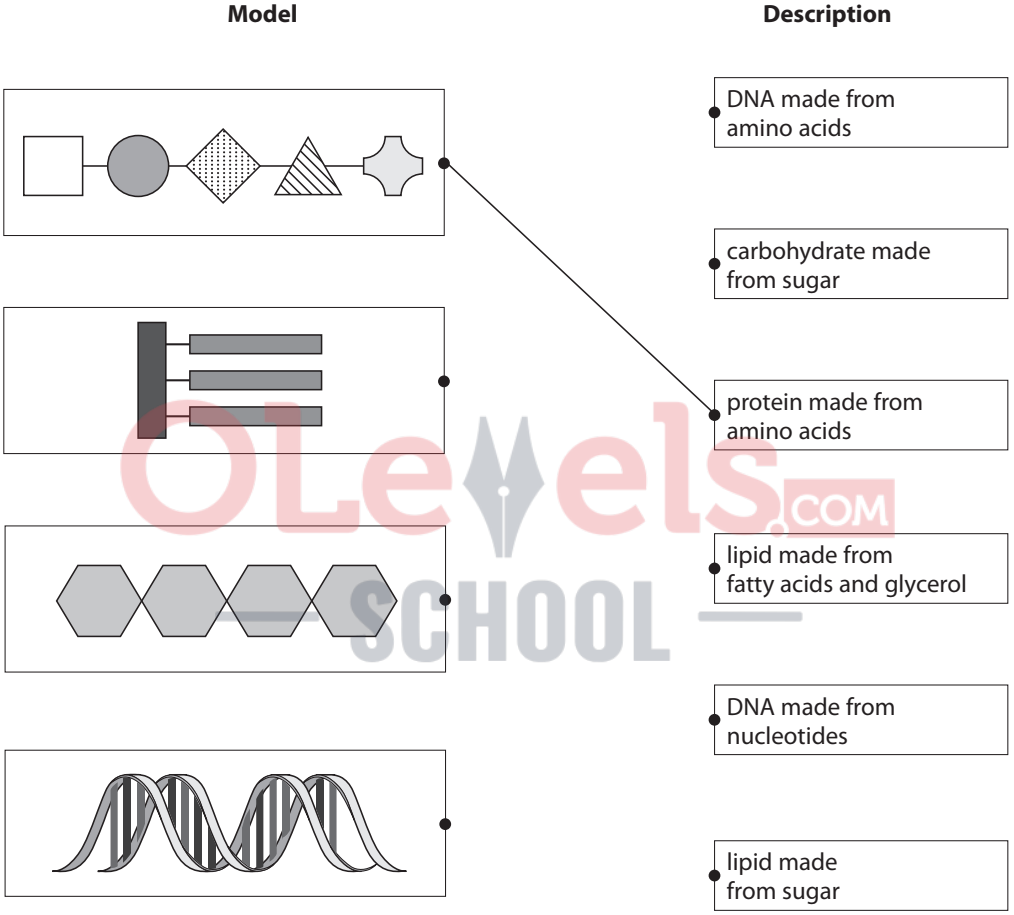
Further practice · Attempt part d only.

**3** (a) The diagram shows models of four biological molecules.

Draw a straight line from each model to its correct description.

One has been done for you.

**(3)**




(b) Which body organ produces enzymes that break down protein?

**(1)**

- ☒ **A** gall bladder
- ☒ **B** large intestine
- ☒ **C** mouth
- ☒ **D** stomach

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## 2. Question 3 · part d · 2 marks

June 2019 Paper 2

Further practice · Attempt part d only.

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(c) A student uses this method to test different substances for protein.

- grind solid substances into small pieces
- place substances into separate test tubes
- add a few drops of Biuret reagent to each substance
- record the colour change for each substance

The table shows the student's results.

| Substance | Colour after Biuret test |
|---|---|
| milk | purple |
| pasta | blue |
| lemon juice | blue |
| cheese | purple |
| distilled water | purple |

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(i) In the student's test, how many of the substances give a positive result for protein?

**(1)**

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(ii) The result for distilled water is incorrect.

Suggest one mistake the student could have made to get this incorrect result.

**(1)**

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(d) The table shows an incomplete risk assessment for the Biuret test.

Complete the table by describing how to reduce the risk of each hazard.

**(2)**

| Hazard | Reducing risk |
|---|---|
| stools – trip hazard | keep stools under bench |
| broken glass – cuts | |
| Biuret reagent – irritant | |

**(Total for Question 3 = 8 marks)**

## 2. Question 3 · part d · 2 marks · Working and answer

Working and answer

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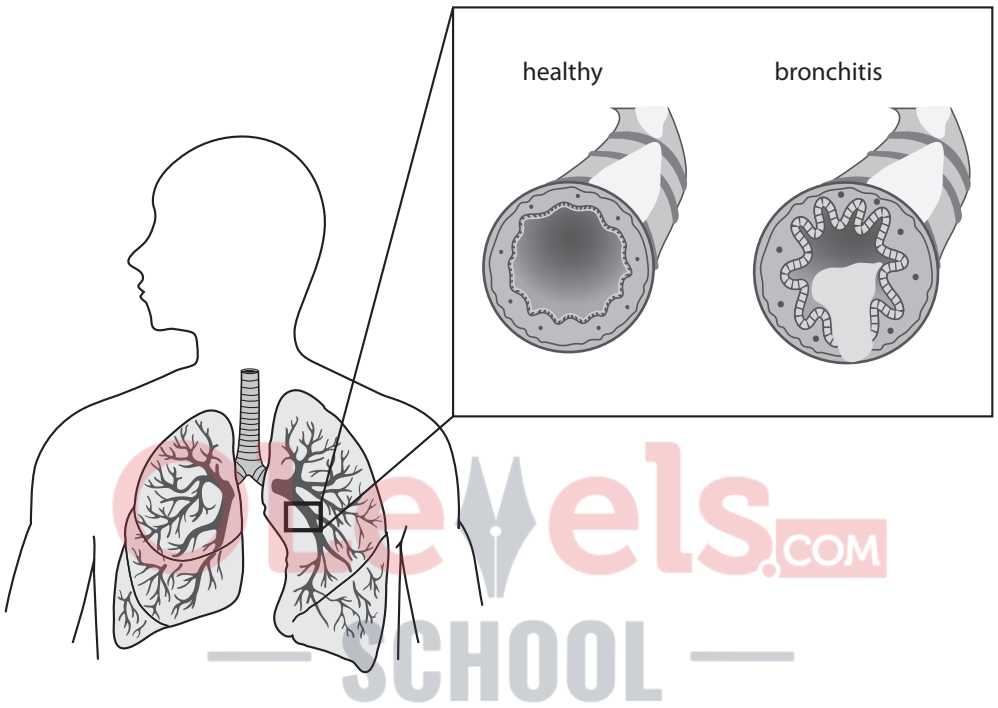
## 3. Question 6 · part b.ii · 1 mark

January 2020 Paper 1

Further practice · Attempt part b.ii only.

**6** Chronic obstructive pulmonary disease (COPD) is a group of lung diseases that includes bronchitis and emphysema.

The diagram shows how bronchitis affects the lungs.




(a) Name the part of the lungs affected by bronchitis.

**(1)**

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## 3. Question 6 · part b.ii · 1 mark

January 2020 Paper 1

Further practice · Attempt part b.ii only.

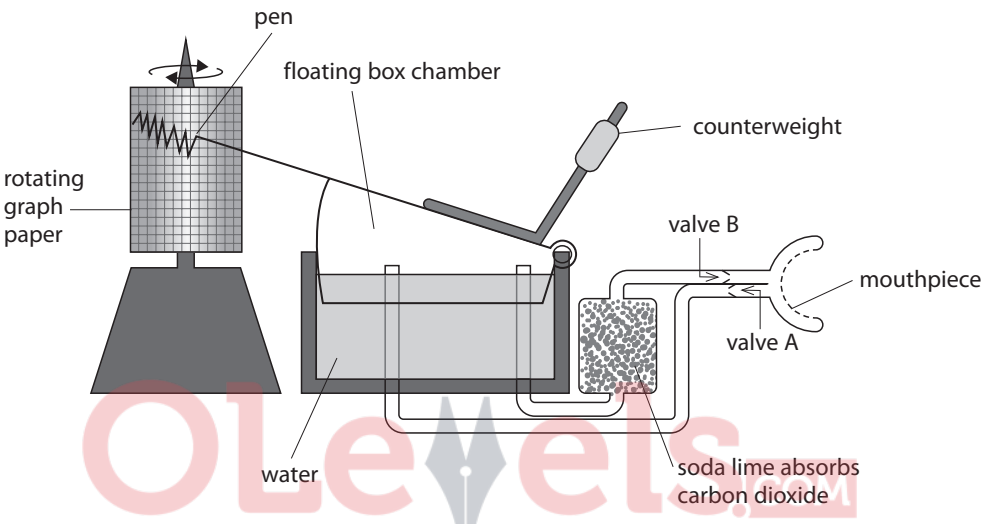
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(b) The diagram shows a spirometer.

A spirometer measures how quickly a person breathes in and out.




(i) Describe how a spirometer is used to obtain a trace of a person's breathing pattern.

**(4)**

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## 3. Question 6 · part b.ii · 1 mark

January 2020 Paper 1

Further practice · Attempt part b.ii only.

(ii) State one way to reduce risk to health when using a spirometer.

**(1)**

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(iii) Explain why a person with bronchitis needs to take a longer breath to obtain a normal tidal volume.

**(2)**

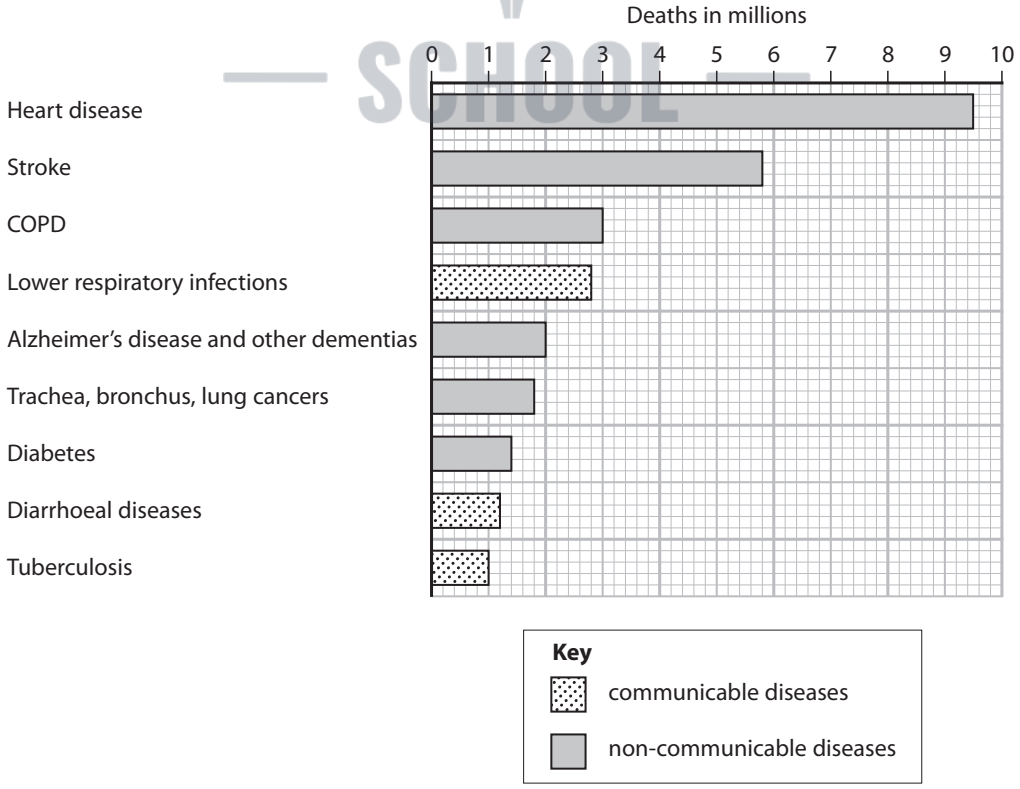
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(c) The bar chart shows the number of deaths in millions from different health conditions worldwide in 2016.




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## 3. Question 6 · part b.ii · 1 mark · Working and answer

Working and answer

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# 4. Question 3 · part a.iv · 2 marks

November 2020 Paper 1R

Further practice · Attempt part a.iv only.

**3** (a) A student uses a light microscope to observe some cells taken from the inside of her cheek.

The table lists steps in the student's method.

The steps are not in the correct order.

(i) Place a number in each box to show the correct order of steps that the student should follow.

The first and last steps have been done for you.

**(2)**

| | **Steps** |
|---|---|
| | select a suitable objective lens |
| 5 | adjust focusing wheel to obtain a clear image |
| 1 | remove cells from the inside of the cheek |
| | place microscope slide onto the microscope stage |
| | smear the cell sample onto a microscope slide |

(ii) The objective lens magnifies cells on the slide making them look bigger.

State an advantage of magnifying the cells.

**(1)**

.....

.....

(iii) Suggest why the student was unable to see the cell sample even though the microscope was at the correct magnification and focused.

**(1)**

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(iv) Give two ways that the student could avoid health hazards when working with living cells.

**(2)**

1 .....

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## 4. Question 3 · part a.iv · 2 marks

November 2020 Paper 1R

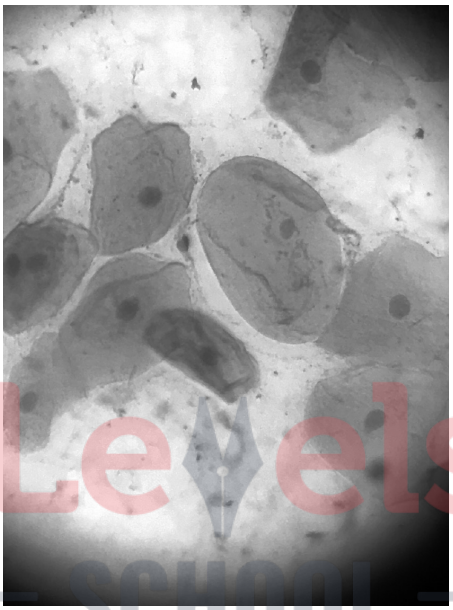
Further practice · Attempt part a.iv only.

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(b) The photograph shows an image of a cheek cell seen under a light microscope using a 10× eyepiece lens and a 40× objective lens.



© Esmiles/Shutterstock

(i) Calculate the total magnification used to view the cheek cell.

**(1)**

total magnification = .....

(ii) The actual diameter of the cheek cell shown in the photograph is 0.018 mm.

Convert the diameter of the cheek cell into micrometre (μm).
[1 μm = $10^{-6}$ m]

**(2)**

diameter = ..... μm

**(Total for Question 3 = 9 marks)**

## 4. Question 3 · part a.iv · 2 marks · Working and answer

Working and answer

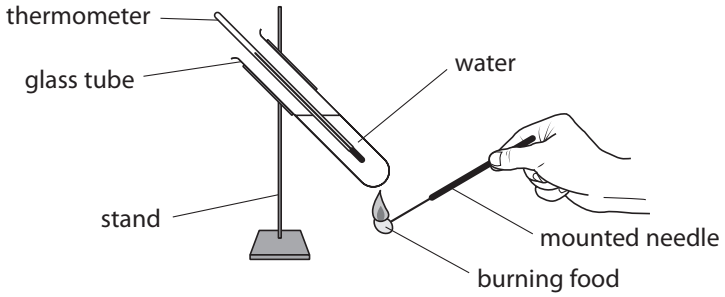
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# 5. Question 7 · part a.i · 3 marks

June 2021 Paper 1

Further practice · Attempt part a.i only.

**7** A student uses this apparatus to investigate the energy content of different foods.




This is the student's method.

- add water to the glass tube
- measure the temperature of the water
- place a piece of food onto the mounted needle
- set the food alight by placing it into a Bunsen flame
- place the burning food under the glass tube
- measure the temperature of the water when the food stops burning

Repeat the method using different foods.

(a) (i) State three safety precautions that the student should take.

**(3)**

1 .....

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2 .....

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3 .....

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P 6 6 4 3 2 A 0 1 8 2 4

## 5. Question 7 · part a.i · 3 marks

June 2021 Paper 1

Further practice · Attempt part a.i only.

(ii) State three factors that should be controlled during the investigation.

**(3)**

1 .....

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2 .....

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3 .....

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(iii) Explain why the energy content of a sample of food is likely to be higher than the energy of the same sample measured by this method.

**(5)**

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## 5. Question 7 · part a.i · 3 marks · Working and answer

Working and answer

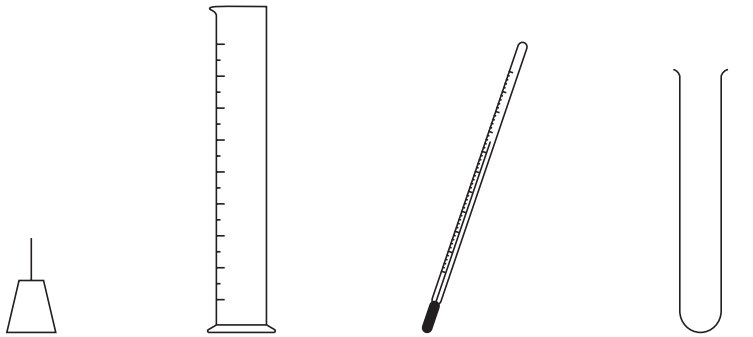
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# 6. Question 3 · part a.iii · 2 marks

June 2022 Paper 1

Further practice · Attempt part a.iii only.

**3** (a) The diagram shows some equipment that can be used to investigate the energy content of different foods.



mounted needle | measuring cylinder | thermometer | test tube

(i) Describe how the equipment can be used to investigate the energy content of different foods.

**(5)**

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## 6. Question 3 · part a.iii · 2 marks

June 2022 Paper 1

Further practice · Attempt part a.iii only.

(ii) Name two variables that need to be controlled in the investigation.

**(2)**

1 .....

.....

2 .....

.....

(iii) Give two safety precautions needed for this investigation.

**(2)**

1 .....

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2 .....

.....

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# 6. Question 3 · part a.iii · 2 marks

June 2022 Paper 1

Further practice · Attempt part a.iii only.

(b) Different people have different energy requirements.

The table shows the daily energy requirements in kilocalories (kcal) of males and females of different ages.

**Daily Energy Requirement in kcal**

| Age of adults in years | Males | Females |
|---|---|---|
| 19–24 | 2772 | 2175 |
| 25–34 | 2749 | 2175 |
| 35–44 | 2629 | 2103 |
| 45–54 | 2581 | 2103 |
| 55–64 | 2581 | 2079 |
| 65–74 | 2342 | 1912 |
| 75 and over | 2294 | 1840 |

(Source: adapted from https://propranolols.blogspot.com/2020/02/daily-nutritional-requirements-for_9.html)

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(i) Describe the patterns in the energy requirements for males and for females.

**(2)**

(ii) Which of these gives a reason for the difference between the energy required by 25-year-old males and the energy required by 25-year-old females?

**(1)**

- ☒ **A** females use more muscle mass than males
- ☒ **B** females eat less than males
- ☒ **C** males eat less than females
- ☒ **D** males use more muscle mass than females

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## 6. Question 3 · part a.iii · 2 marks · Working and answer

Working and answer

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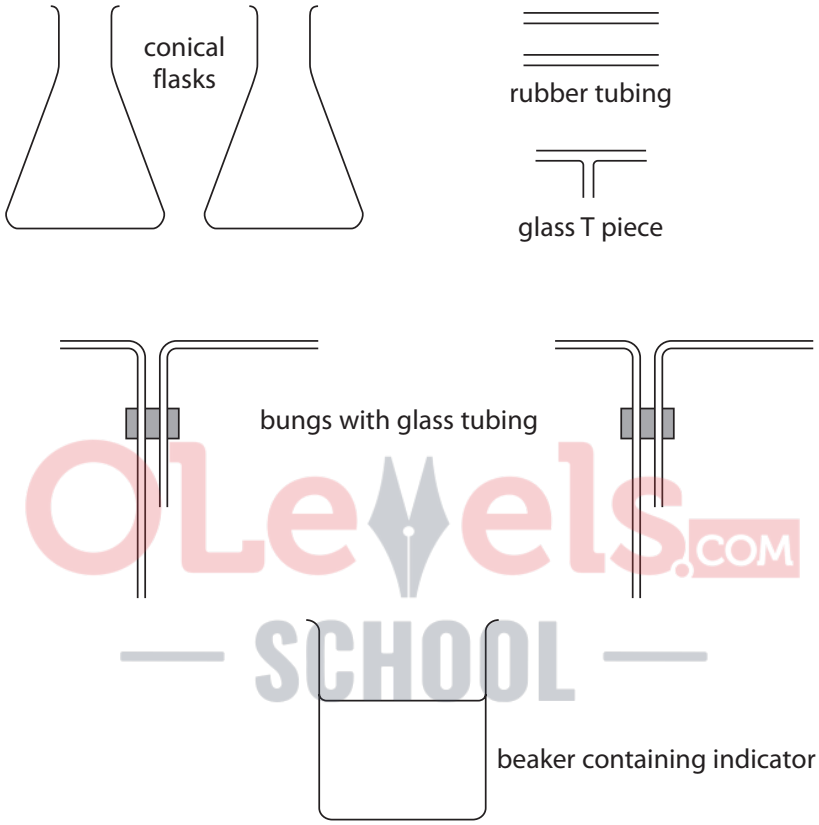
# 7. Question 1 · part d · 1 mark

June 2022 Paper 2R

Further practice · Attempt part d only.

**1** A student investigates whether there is more carbon dioxide in expired air than in inspired air.

This is the apparatus the student uses.




The apparatus is not drawn to scale.

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## 7. Question 1 · part d · 1 mark

June 2022 Paper 2R

Further practice · Attempt part d only.

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(a) Draw a diagram to show how the apparatus should be put together for the investigation.

**(3)**

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(b) Name an indicator that could be used in the investigation.

**(1)**

(c) Describe how the apparatus would be used to determine whether there is more carbon dioxide in expired air than in inspired air.

**(4)**

## 7. Question 1 · part d · 1 mark

June 2022 Paper 2R

Further practice · Attempt part d only.

(d) State one safety precaution.

**(1)**

.....

.....

(e) State one variable that should be controlled.

**(1)**

.....

.....

**(Total for Question 1 = 10 marks)**

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Working and answer

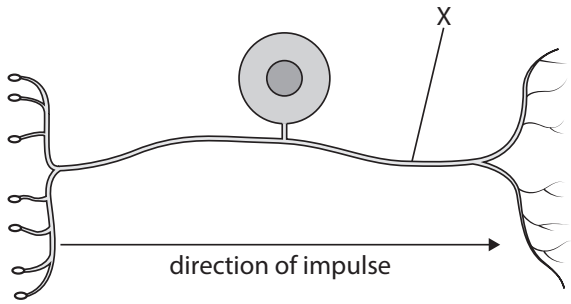
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# 8. Question 1 · part c.i · 1 mark

June 2024 Paper 1

Further practice · Attempt part c.i only.

**1** (a) The diagram shows a cell found in the human nervous system.




(Source: © Designua/ Shutterstock)

(i) The cell shown in the diagram transmits electrical impulses from a receptor cell to the central nervous system.

State the name of this type of nerve cell.

**(1)**

.....

(ii) Name the structure labelled X.

**(1)**

.....

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# 8. Question 1 · part c.i · 1 mark

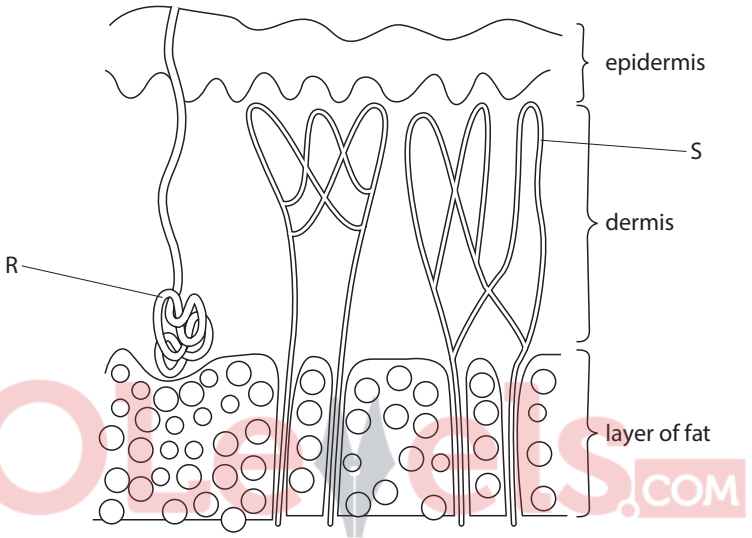
June 2024 Paper 1

Further practice · Attempt part c.i only.

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(b) Structures in the skin respond to information from the nervous system.

The diagram shows a section of the skin.




(Source: adapted from Human Physiology and Health, Author David Wright, Publisher Heinemann page 113)

(i) Name part R and part S.

**(2)**

part R

.....

part S

.....

(ii) Describe how part R helps to cool the body.

**(2)**

.....

.....

.....

.....

## 8. Question 1 · part c.i · 1 mark

June 2024 Paper 1

Further practice · Attempt part c.i only.

(iii) Describe the role of the layer of fat in maintaining body temperature.

**(2)**

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## 8. Question 1 · part c.i · 1 mark

June 2024 Paper 1

Further practice · Attempt part c.i only.

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(c) A student investigates the effect of temperature on receptors in the skin.

This is their method.

- place one hand in hot water
- place the other hand in ice cold water
- after 2 minutes take both hands out of the water
- place both hands in water at room temperature

(i) State one possible hazard with their method.

**(1)**

.....

.....

(ii) Describe how to reduce the risk of this hazard.

**(2)**

.....

.....

.....

.....

(iii) When the student placed both hands in the water at room temperature the water felt hot to one hand and ice cold to the other.

Name the structure in the skin that detects temperature.

**(1)**

.....

.....

**(Total for Question 1 = 12 marks)**

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## 8. Question 1 · part c.i · 1 mark · Working and answer

Working and answer

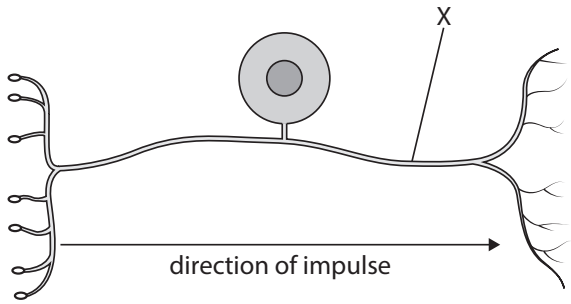
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# 9. Question 1 · part c.ii · 2 marks

June 2024 Paper 1

Further practice · Attempt part c.ii only.

**1** (a) The diagram shows a cell found in the human nervous system.




(Source: © Designua/ Shutterstock)

(i) The cell shown in the diagram transmits electrical impulses from a receptor cell to the central nervous system.

State the name of this type of nerve cell.

**(1)**

.....

(ii) Name the structure labelled X.

**(1)**

.....

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P 7 4 6 2 6 R A 0 2 2 4

# 9. Question 1 · part c.ii · 2 marks

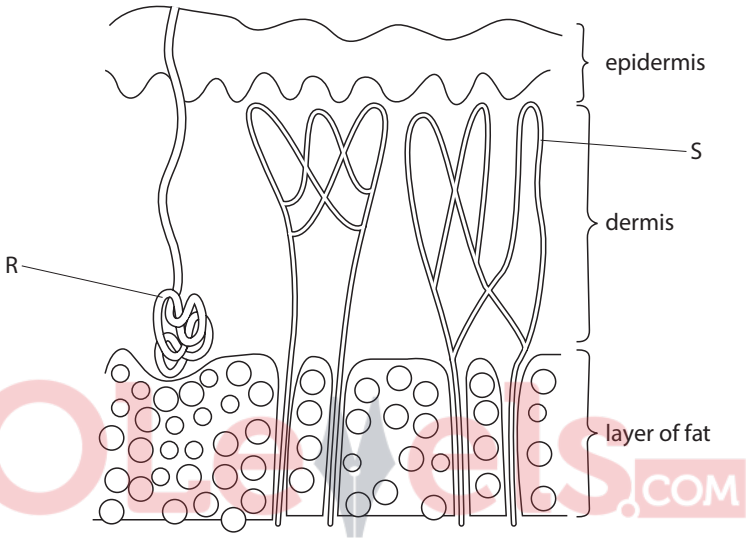
June 2024 Paper 1

Further practice · Attempt part c.ii only.

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(b) Structures in the skin respond to information from the nervous system.

The diagram shows a section of the skin.




(Source: adapted from Human Physiology and Health, Author David Wright, Publisher Heinemann page 113)

(i) Name part R and part S.

**(2)**

part R

.....

part S

.....

(ii) Describe how part R helps to cool the body.

**(2)**

.....

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## 9. Question 1 · part c.ii · 2 marks

June 2024 Paper 1

Further practice · Attempt part c.ii only.

(iii) Describe the role of the layer of fat in maintaining body temperature.

**(2)**

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## 9. Question 1 · part c.ii · 2 marks

June 2024 Paper 1

Further practice · Attempt part c.ii only.

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(c) A student investigates the effect of temperature on receptors in the skin.

This is their method.

- place one hand in hot water
- place the other hand in ice cold water
- after 2 minutes take both hands out of the water
- place both hands in water at room temperature

(i) State one possible hazard with their method.

**(1)**

.....

.....

(ii) Describe how to reduce the risk of this hazard.

**(2)**

.....

.....

.....

.....

(iii) When the student placed both hands in the water at room temperature the water felt hot to one hand and ice cold to the other.

Name the structure in the skin that detects temperature.

**(1)**

.....

.....

**(Total for Question 1 = 12 marks)**

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## 9. Question 1 · part c.ii · 2 marks · Working and answer

Working and answer

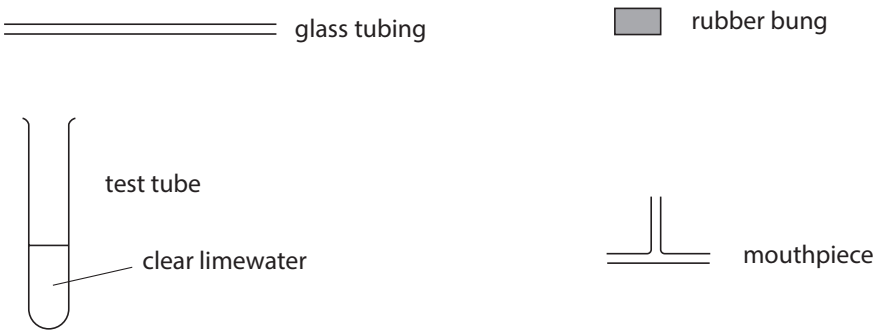
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# 10. Question 4 · part a.ii · 1 mark

June 2024 Paper 1

Further practice · Attempt part a.ii only.

**4** (a) The diagram shows some scientific equipment.




(i) Draw a diagram to show how this equipment is set up to investigate the difference between the carbon dioxide concentration of inspired air and of expired air.

**(3)**

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# 10. Question 4 · part a.ii · 1 mark

June 2024 Paper 1

Further practice · Attempt part a.ii only.

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(ii) State one safety precaution in this investigation.

**(1)**

(b) A student investigates their breathing rate at rest, after gentle exercise and again after vigorous exercise.

The student repeats each exercise three times.

The student's mean results are shown in the table.

| Mean breathing rate in breaths per minute | | |
|---|---|---|
| **at rest** | **after gentle exercise** | **after vigorous exercise** |
| 14 | 19 | 50 |

(i) Give one control variable in this investigation.

**(1)**

(ii) Explain how the breathing rate after vigorous exercise benefits working muscles.

**(4)**

**(Total for Question 4 = 9 marks)**

## 10. Question 4 · part a.ii · 1 mark · Working and answer

Working and answer

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# 11. Question 8 · part b.ii · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.ii only.

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**8** (a) Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

The human body has physical and chemical barriers that reduce the risk of pathogens entering the body. These barriers are known as the first line of defence. If the barriers are penetrated, then the second line of defence comes into action.

The cells involved in the second line of defence form part of the non-specific line of defence. They do not target specific pathogens but they play an important role in stimulating cells involved in the next stages of defence against disease.

The third line of defence involves specific immune responses by cells. These cells rely on the presence and detection of marker chemicals found on the surface of microscopic invaders. The cells of the specific immune system have receptor proteins that interact with the marker chemicals. The marker chemicals cause the cells to become activated and produce antibodies. The production of antibodies in this way is known as natural, active immunity and results in long-term immunity. However, there are other ways that antibodies can provide protection against disease. Natural passive immunity and artificial passive immunity are two similar processes that minimise the risk of too much harm being caused by pathogens in the body.

(i) Name a physical barrier that prevents the entry of pathogens into the body. (lines 1 to 3)

**(1)**

(ii) Suggest how chemical barriers can reduce the harm caused to the body by pathogens. (lines 1 to 3)

**(1)**

(iii) Phagocytes are one type of cell involved in the second line of defence.

Describe the process of phagocytosis. (lines 4 to 6)

**(2)**

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# 11. Question 8 · part b.ii · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.ii only.

(iv) State the scientific name for the marker chemicals found on the surface of microscopic invaders. (lines 7 to 11)

**(1)**

(v) Suggest what is meant by a specific immune response. (line 7)

**(2)**

(vi) Describe the differences between natural passive immunity and artificial passive immunity. (lines 14 to 16)

You should give examples in your answer.

**(2)**

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# 11. Question 8 · part b.ii · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.ii only.

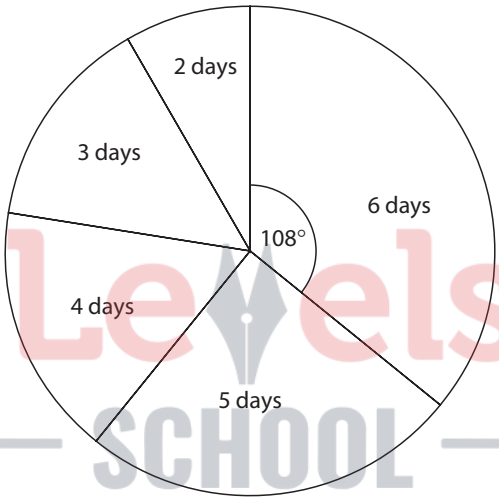
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(b) Scientists test how quickly 30 different types of antibiotic reduce the population size of a resistant strain of bacteria.

The scientists record the number of antibiotics that reduce the population of the resistant strain of bacteria by half.

They do this after 2 days, 3 days, 4 days, 5 days or 6 days.

Each section of the pie chart represents the number of antibiotics.




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(i) Calculate the number of types of antibiotic that reduce the population size of the resistant strain of bacteria by half in 6 days.

**(3)**

number of types of antibiotic = .....



P 7 4 6 3 2 A 0 2 3 2 4

*Turn over* ▶

# 11. Question 8 · part b.ii · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.ii only.

(ii) State a reason why it is important to wear gloves and a face mask when investigating bacteria.

**(1)**

(iii) Describe how the use of antibiotics can cause antibiotic-resistant strains of bacteria.

**(2)**

**(Total for Question 8 = 15 marks)**

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Working and answer

# Practice workbook · Improve reliability with repeats, means and larger samples

Investigations done once or on few individuals; asks how to make results more reliable, why many people are used, or how to treat an anomaly.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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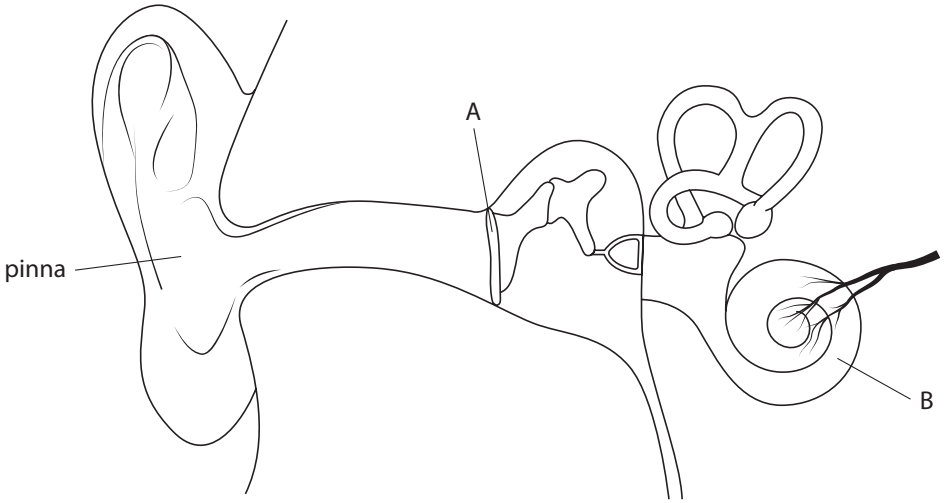
Type ID: HBI-PRAC-IMPROVE-RELIABILITY-REPEATS-SAMPLE

# 1. Question 5 · part c.ii · 2 marks

June 2022 Paper 2

Worked-example retry · Attempt part c.ii only.

**5** The diagram shows a section through the human ear.




(a) (i) Name parts A and B. **(2)**

A .....

B .....

(ii) Describe the function of the pinna. **(2)**

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# 1. Question 5 · part c.ii · 2 marks

June 2022 Paper 2

Worked-example retry · Attempt part c.ii only.

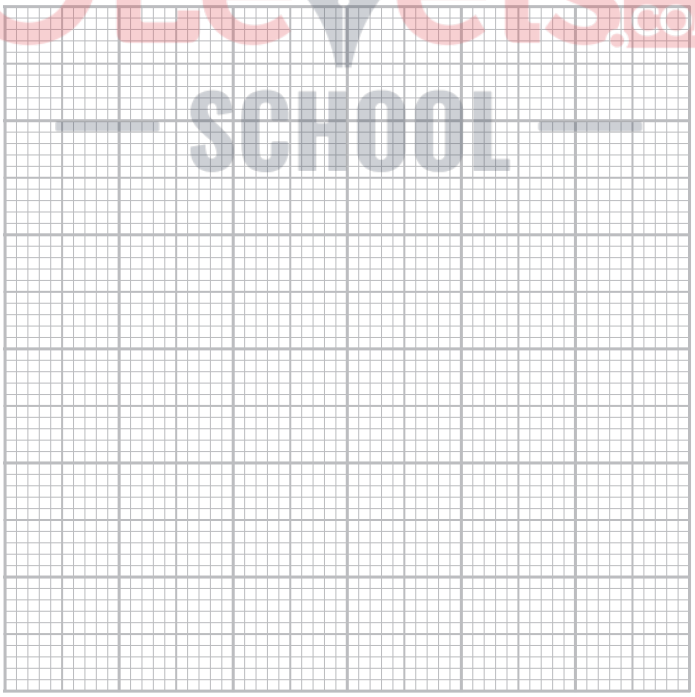
(b) The range of frequencies audible to the human ear can vary with age.

The table shows the maximum frequency detected by people of different ages.

| Age in years | Maximum frequency detected in kHz |
|---|---|
| 20 | 20 |
| 30 | 18 |
| 40 | 16 |
| 50 | 14 |

(i) Draw a bar chart to show how the maximum frequency detected varies with age.

**(4)**



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# 1. Question 5 · part c.ii · 2 marks

June 2022 Paper 2

Worked-example retry · Attempt part c.ii only.

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(ii) Calculate the percentage change in the maximum frequency detected by people aged 20 and people aged 50.

**(3)**

percentage change = ..... %

(c) A student suggests this method that could be used to produce the data shown in the table.

- select a person of each age to be tested
- use a signal generator to produce sounds at different frequencies
- record the maximum frequency detected by each person

(i) Give two reasons why this is unlikely to produce valid results.

**(2)**

1 .....

.....

2 .....

.....

(ii) Describe how the reliability of the results could be improved.

**(2)**

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**(Total for Question 5 = 15 marks)**

## 1. Question 5 · part c.ii · 2 marks · Working and answer

Working and answer

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## 2. Question 5 · part a.iii · 2 marks

June 2019 Paper 1

Further practice · Attempt part a.iii only.

**5** A teacher wants to calculate the body mass index (BMI) of a number of students.

(a) The teacher measures the height of students from four age groups.

There are two male and two female students in each age group.

The table shows the data collected.

| Age in years | Height of student in cm | | | | Mean height of students in cm |
|---|---|---|---|---|---|
| | Male | | Female | | |
| 12 | 146.2 | 148.9 | 142.8 | 144.2 | 145.5 |
| 14 | 165.7 | 166.4 | 161.9 | 164.0 | 164.5 |
| 16 | 175.9 | 178.5 | 166.3 | 167.6 | |
| 18 | 180.9 | 181.3 | 171.2 | 174.3 | 176.9 |

(i) Calculate the mean height of the students aged 16.

Give your answer to one decimal place.

**(2)**

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mean height = ..... cm

(ii) State two conclusions that the teacher could make from the data in the table.

**(2)**

1 .....

.....

2 .....

.....

(iii) Give two ways the teacher could improve his investigation.

**(2)**

1 .....

.....

2 .....

.....

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## 2. Question 5 · part a.iii · 2 marks

June 2019 Paper 1

Further practice · Attempt part a.iii only.

(b) The table shows data for two different students.

| Student | Height in m | Mass in kg | BMI |
|---|---|---|---|
| X | 1.69 | 62 | 21.7 |
| Y | 1.46 | 71 | |

(i) BMI is calculated using the equation

$$\text{BMI} = \frac{\text{mass}}{\text{height}^2}$$

Calculate the BMI of student Y.
[mass measured in kg, height measured in m]

**(2)**

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BMI of student Y = .....

(ii) BMI can be used to assess whether a person has a healthy mass for their height.

| BMI | Classification |
|---|---|
| below 18.5 | underweight |
| 18.5 – 24.9 | normal weight |
| 25 – 29.9 | overweight |
| 30 – 40 | obese |
| above 40 | morbidly obese |

Using information from the table, explain how the BMI of student Y could affect her health.

**(2)**

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**(Total for Question 5 = 10 marks)**

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## 2. Question 5 · part a.iii · 2 marks · Working and answer

Working and answer

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# 3. Question 5 · part a.ii · 2 marks

November 2020 Paper 2R

Further practice · Attempt part a.ii only.

**5** A scientist investigates how the drug, atropine, affects the heart rate in humans.

Atropine paralyses the vagus nerve which carries impulses from the brain to the heart.

The heart rates of ten people are measured at intervals of two minutes for eight minutes, after which, atropine is injected.

The scientist continues to measure the heart rates.

The scientist's results are shown in the table.

| Time in minutes | Heart rate in beats per minute |
|---|---|
| 0 | 90 |
| 2 | 81 |
| 4 | 78 |
| 6 | 80 |
| 8 | 82 |
| 9 | 100 |
| 10 | 120 |
| 11 | 148 |
| 12 | 160 |
| 13 | 138 |
| 15 | 128 |
| 17 | 120 |
| 19 | 116 |
| 21 | 109 |
| 23 | 100 |

atropine injected →

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P 6 2 0 5 4 A 0 1 2 2 4

## 3. Question 5 · part a.ii · 2 marks

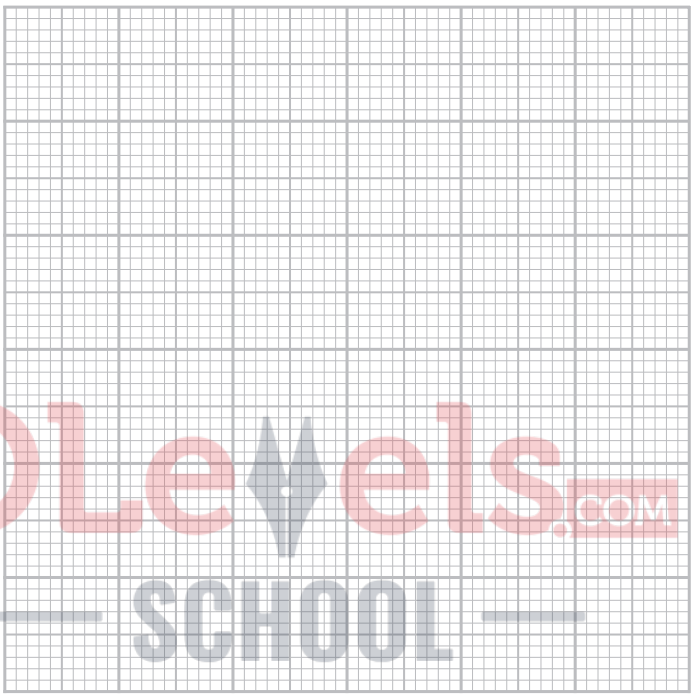
November 2020 Paper 2R

Further practice · Attempt part a.ii only.

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(a) (i) Plot the scientist's results on the grid. Draw a curve of best fit.

**(6)**



(ii) Explain why ten people are injected with atropine rather than just one person.

**(2)**

.....

.....

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.....

(b) (i) Calculate the percentage change in the heart rate at 12 minutes compared to the heart rate at the start of the investigation.

**(2)**

percentage change = .....%

## 3. Question 5 · part a.ii · 2 marks · Working and answer

Working and answer

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## 4. Question 7 · part b.iii · 1 mark

November 2020 Paper 2R

Further practice · Attempt part b.iii only.

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**7** Chymosin is an enzyme found in young humans, but not in adults. It converts soluble proteins in milk into solid proteins.

(a) (i) Suggest the advantages to a young human of having chymosin in their alimentary canal.

**(2)**

(ii) Explain why chymosin is only needed in young humans.

**(2)**

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(iii) Describe a test to show that the solid formed by the action of chymosin on milk is a protein.

**(3)**

# 4. Question 7 · part b.iii · 1 mark

November 2020 Paper 2R

Further practice · Attempt part b.iii only.

(b) A student investigates the effect of carbon dioxide on the activity of chymosin.

The student bubbles different volumes of carbon dioxide gas into five samples of milk.

He then adds chymosin to each sample and records the time taken for the milk protein to become solid.

The student repeats this method three times.

The table shows the student's results.

| Number of bubbles of $CO_2$ | Time taken for chymosin to turn milk protein solid, in seconds | | | | |
|---|---|---|---|---|---|
| | Test 1 | Test 2 | Test 3 | Test 4 | Mean |
| 100 | 253 | 257 | 250 | 260 | 255 |
| 150 | 238 | 232 | 241 | 229 | |
| 200 | 216 | 214 | 219 | 211 | 215 |
| 250 | 208 | 202 | 212 | 198 | 205 |
| 300 | 210 | 200 | 199 | 311 | 203 |

(i) State three factors that the student should control.

**(3)**

1 .....

.....

2 .....

.....

3 .....

.....

(ii) Calculate the missing mean (average) time taken for 150 bubbles.

**(2)**

mean time taken = .....

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**20**

P 6 2 0 5 4 A 0 2 0 2 4

# 4. Question 7 · part b.iii · 1 mark

November 2020 Paper 2R

Further practice · Attempt part b.iii only.

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(iii) In test 4 there is an anomalous result.

State how the student deals with this result.

**(1)**

.....

.....

(iv) State why recording the number of bubbles may produce inaccurate results.

**(1)**

.....

.....

**(Total for Question 7 = 14 marks)**

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Working and answer

# 5. Question 1 · part a.ii · 1 mark

November 2021 Paper 1

Further practice · Attempt part a.ii only.

**1** The enzyme catalase is found in the liver.

Catalase acts as a catalyst when added to hydrogen peroxide solution, causing the solution to break down into water and oxygen.

A student investigates the increase in the rate of breakdown of hydrogen peroxide into water and oxygen using catalase.

This is the student's method.

- put 100 g of liver into 100 $cm^3$ of water, and mix into a smooth paste
- put 10 $cm^3$ of the paste into a beaker at room temperature
- add one drop of hydrogen peroxide to the paste
- assess how many oxygen bubbles are produced on a scale of 0 to 5, with 5 representing most bubbles and 0 representing no bubbles

The student repeats the investigation using different conditions.

The conditions he uses are:

- liver paste heated to 70 °C
- liver paste that has been kept at 3 °C for an hour
- acidified liver paste
- liver paste with an alkali added
- liver paste with no hydrogen peroxide added

These are the student's results.

*room temperature = 5  heated to 70 °C = 0*

*kept at 3 °C = 1  acid added = 2  alkali added = 3*

*no hydrogen peroxide added = 0*

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# 5. Question 1 · part a.ii · 1 mark

November 2021 Paper 1

Further practice · Attempt part a.ii only.

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(a) (i) Give the student's results in the form of a suitable table.

**(3)**

(ii) State why this investigation is not reliable.

**(1)**

(iii) Give a reason why the results of this investigation are not accurate.

**(1)**

(iv) Describe how the investigation could be improved so that the conclusions could be more accurate and reliable.

**(3)**

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## 5. Question 1 · part a.ii · 1 mark · Working and answer

Working and answer

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# 6. Question 2 · part b.iii · 2 marks

June 2023 Paper 2R

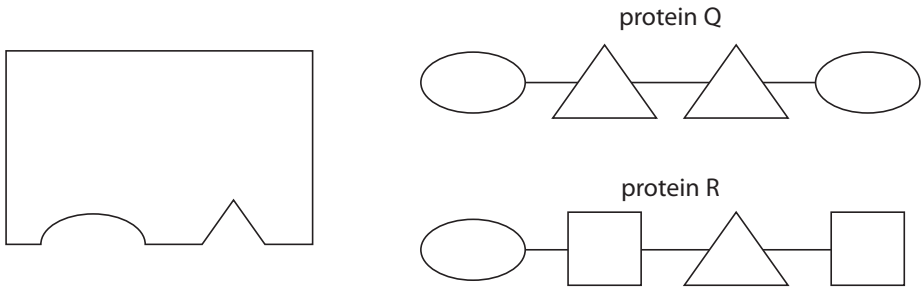
Further practice · Attempt part b.iii only.

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**2** (a) The diagram represents an enzyme, known as enzyme P, and two named substrates.




**Enzyme P** **Substrates**

(i) Use the diagram and words from the box to complete the passage about enzyme P.

**(3)**

| | | | | | | |
|---|---|---|---|---|---|---|
| amino acids | amylases | fatty acids | proteases | protein Q | protein R | starch | sugar |

Enzyme P belongs to a group of enzymes called ..... .

This enzyme breaks down ..... in food

to ..... .

(ii) Give one location where enzyme P is found in the digestive system.

**(1)**

.....

(iii) Name the molecule that codes for the production of enzyme P.

**(1)**

.....

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# 6. Question 2 · part b.iii · 2 marks

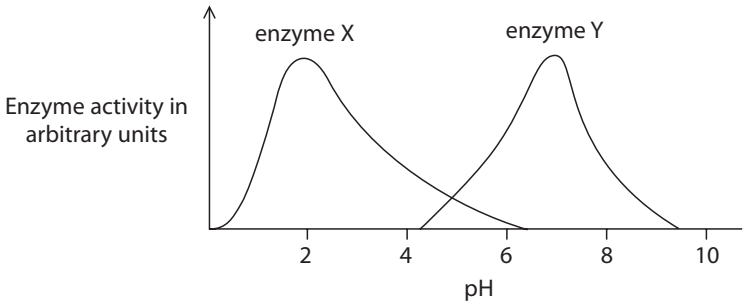
June 2023 Paper 2R

Further practice · Attempt part b.iii only.

(b) A student investigates the effect of pH on the activity of two enzymes, X and Y.

The student measures the activity of each enzyme once, using different pH values.

The graph shows the results of their investigation.




(i) Describe how the activity of enzyme X changes with pH.

**(3)**

(ii) Explain why no activity is shown for enzyme Y at pH2.

**(2)**

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P 7 2 5 8 8 A 0 6 2 4

## 6. Question 2 · part b.iii · 2 marks

June 2023 Paper 2R

Further practice · Attempt part b.iii only.

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(iii) Describe how the student could make the results of this investigation more reliable.

**(2)**

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(iv) Name one control variable in the student's investigation.

**(1)**

.....

**(Total for Question 2 = 13 marks)**

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Working and answer

## 7. Question 4 · part b.i · 2 marks

June 2023 Paper 2

Further practice · Attempt part b.i only.

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**4** (a) (i) Complete the word equation for one type of respiration.

**(2)**

oxygen + ..... → ..... + .....

(ii) State the type of respiration shown by the equation.

**(1)**

.....

(iii) In which part of the cell does this type of respiration occur?

**(1)**

- ☒ **A** endoplasmic reticulum
- ☒ **B** mitochondrion
- ☒ **C** nucleus
- ☒ **D** ribosome

(iv) Name the molecule used to transfer energy in a cell.

**(1)**

.....

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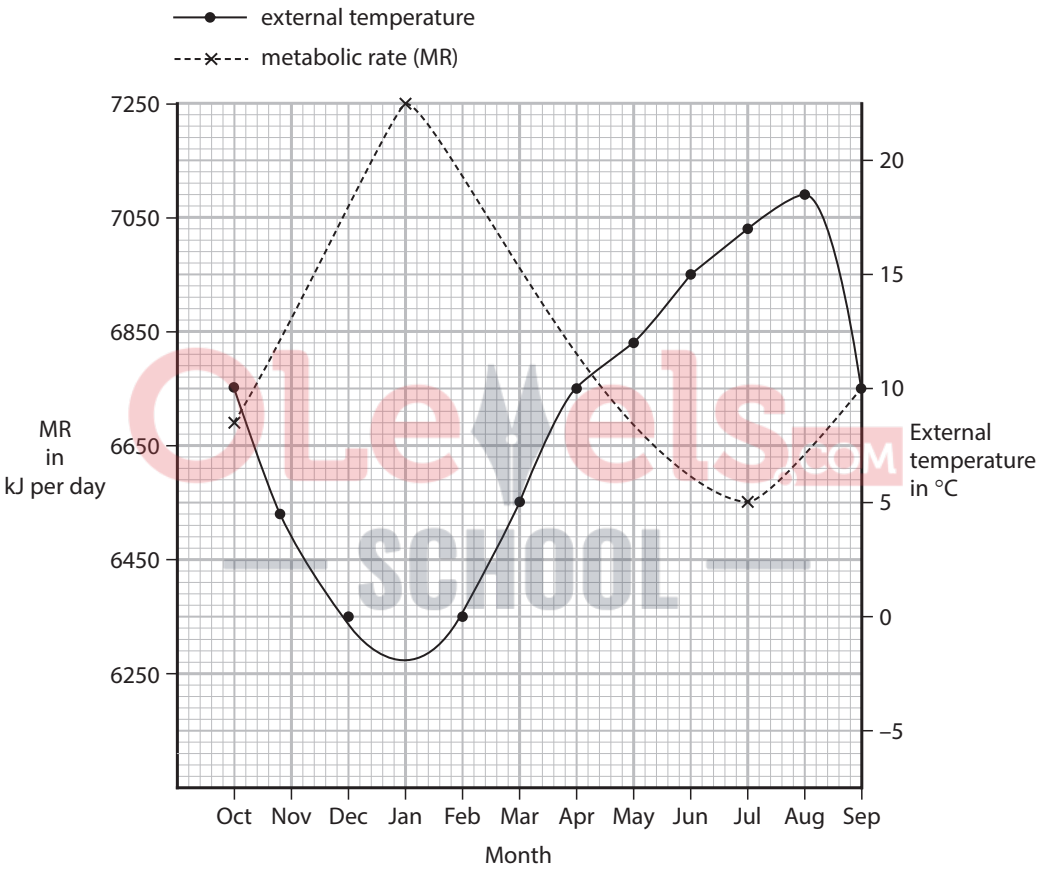
# 7. Question 4 · part b.i · 2 marks

June 2023 Paper 2

Further practice · Attempt part b.i only.

(b) An investigation was carried out to see how a person's metabolic rate (MR) and the external temperature change at different times in one year.

The graph shows the results of the investigation for one person.




(i) Describe how the investigation could be made more reliable.

**(2)**

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P 7 2 5 8 7 A 0 1 0 2 4

## 7. Question 4 · part b.i · 2 marks

June 2023 Paper 2

Further practice · Attempt part b.i only.

(ii) Determine the difference between the maximum metabolic rate and minimum metabolic rate.

**(2)**

difference = ..... kJ per day

(iii) Explain why the metabolic rate changes throughout the year.

**(5)**

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## 7. Question 4 · part b.i · 2 marks · Working and answer

Working and answer

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# 8. Question 2 · part e · 1 mark

November 2024 Paper 1

Further practice · Attempt part e only.

**2** A student investigates the activity of an enzyme on fat found in milk.
This is the student's method.

- pour milk and enzyme solution into a test tube
- determine the pH of the solution every minute for 10 minutes

The table shows the student's results.

| Time in minutes | pH of solution |
|---|---|
| 1 | 6.8 |
| 2 | 6.4 |
| 3 | 6.0 |
| 4 | 5.8 |
| 5 | 5.6 |
| 6 | 5.2 |
| 7 | 5.0 |
| 8 | 4.8 |
| 9 | 4.6 |
| 10 | 4.6 |

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## 8. Question 2 · part e · 1 mark

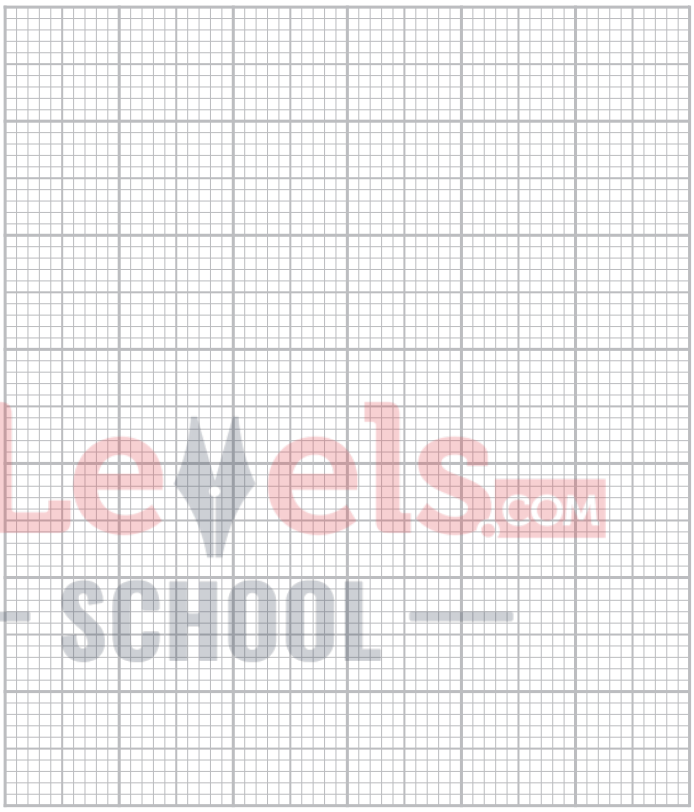
November 2024 Paper 1

Further practice · Attempt part e only.

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(a) (i) Plot a graph of the student's results.

**(3)**



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(ii) Draw a line of best fit.

**(1)**

(iii) Give the trend shown by the graph.

**(1)**

.....

.....

## 8. Question 2 · part e · 1 mark

November 2024 Paper 1

Further practice · Attempt part e only.

(b) Explain the difference in the pH at 1 minute and the pH at 10 minutes.

**(3)**

(c) Name the independent variable in this investigation.

**(1)**

(d) Give two variables that the student should control in this investigation.

**(2)**

1 .....

2 .....

(e) State how the student could improve the reliability of their results.

**(1)**

**(Total for Question 2 = 12 marks)**

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## 8. Question 2 · part e · 1 mark · Working and answer

Working and answer

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# 9. Question 7 · part d · 2 marks

June 2021 Paper 2

Further practice · Attempt part d only.

**7** Garlic plants contain a substance called allicin. This substance is thought to have antibacterial properties.

A student investigates the antibacterial effect of garlic.

This is the student's method.

- label three test tubes A, B and C
- add 10 $cm^3$ of milk into each of the tubes
- add a sample of the bacterium *E. coli* into tubes B and C using aseptic techniques
- crush the garlic in water to form garlic juice
- add 1 $cm^3$ of garlic juice to tube C
- shake all three tubes and leave to incubate for two hours
- place a sample from tube A into the centre of Petri dish A, from tube B into the centre of Petri dish B and from tube C into the centre of Petri dish C
- incubate the three Petri dishes for five days
- each day measure the diameter of any bacterial growth that has occurred

The table shows the student's results.

| Petri dish | Diameter of bacterial growth in mm | | | | |
|---|---|---|---|---|---|
| | day 1 | day 2 | day 3 | day 4 | day 5 |
| A | 0 | 1.7 | 3.0 | 4.6 | 7.1 |
| B | 0 | 4.2 | 8.4 | 15.1 | 36.5 |
| C | 0 | 0 | 0 | 0 | 0 |

(a) (i) State three control variables in the student's investigation.

**(3)**

1 .....

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2 .....

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3 .....

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# 9. Question 7 · part d · 2 marks

June 2021 Paper 2

Further practice · Attempt part d only.

(ii) State the dependent variable in the student's investigation.

**(1)**

.....

(b) (i) Calculate the mean daily increase in the diameter of the bacterial growth for dish A and dish B.

**(2)**

mean daily increase for dish A ..... mm/day

mean daily increase for dish B ..... mm/day

(ii) Explain the difference between the mean daily increases in the diameter of the bacterial growth for dish A and dish B.

**(4)**

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(c) Explain what is meant by an aseptic techniques for the transfer of bacteria.

**(2)**

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# 9. Question 7 · part d · 2 marks

June 2021 Paper 2

Further practice · Attempt part d only.

(d) Suggest what further tests the student should do to show that garlic can be used as an effective general antibacterial agent.

**(2)**

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**(Total for Question 7 = 14 marks)**

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Working and answer

# Practice workbook · State the variables to control in an investigation

Bench investigations and investigations on people where the student lists factors to keep the same.

21 question parts: 1 worked-example retry and 20 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-PRAC-NAME-CONTROL-VARIABLES

# 1. Question 2 · part a.ii · 3 marks

June 2022 Paper 2R

Worked-example retry · Attempt part a.ii only.

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**2** A student investigates the effect of a competitive inhibitor on the action of the enzyme phosphatase.

This is the reaction catalysed by the enzyme.

phenolphthalein phosphate ⟶ phenolphthalein + phosphate

Phenolphthalein turns pink when an alkali is added.

This colour can be measured using a colorimeter. The more intense the colour, the higher the reading on the colorimeter.

This is the student's method.

- prepare six different concentrations of phenolphthalein phosphate solution
- place 5.0 $cm^3$ of each solution into a different test tube
- add 1.0 $cm^3$ of a competitive inhibitor and 1.0 $cm^3$ of the enzyme phosphatase into each test tube
- wait 30 minutes, then add 1.0 $cm^3$ of an alkali to each tube
- measure the intensity of the pink colour using a colorimeter

(a) (i) Give a suitable control for this investigation. **(2)**

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(ii) State three variables that should be kept constant in this investigation. **(3)**

1 .....

2 .....

3 .....

# 1. Question 2 · part a.ii · 3 marks

June 2022 Paper 2R

Worked-example retry · Attempt part a.ii only.

(b) The table shows the student's results.

| Concentration of phenolphthalein phosphate solution in arbitrary units | Intensity of pink colour in arbitrary units |
|---|---|
| 0.05 | 0.20 |
| 0.10 | 0.30 |
| 0.20 | 0.47 |
| 0.40 | 0.65 |
| 0.60 | 0.79 |
| 0.80 | 0.91 |

(i) Plot the results of the investigation. (3)

(ii) Draw a curve of best fit. (1)

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## 1. Question 2 · part a.ii · 3 marks · Working and answer

Working and answer

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# 2. Question 2 · part c.ii · 2 marks

June 2019 Paper 1R

Further practice · Attempt part c.ii only.

**2** Humans need a balanced diet.

(a) Vitamins are needed in a balanced diet.

State four other components of a balanced diet.

**(2)**

1 .....

2 .....

3 .....

4 .....

(b) Explain why vitamin C is needed in a balanced diet.

**(2)**

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## 2. Question 2 · part c.ii · 2 marks

June 2019 Paper 1R

Further practice · Attempt part c.ii only.

(c) The vitamin C content of fruit juices varies.

(i) Design an investigation to compare the vitamin C content of fresh fruit juice and long-life fruit juice.

**(5)**

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(ii) State two variables that should be controlled in this investigation.

**(2)**

1 .....

2 .....

**(Total for Question 2 = 11 marks)**

## 2. Question 2 · part c.ii · 2 marks · Working and answer

Working and answer

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## 3. Question 9 · part b.i · 1 mark

June 2019 Paper 2

Further practice · Attempt part b.i only.

**9** Aspirin is a painkiller, but can also be prescribed to people who are at risk of having a stroke or a heart attack.

Aspirin dissolves blood clots and is also an enzyme inhibitor, which reduces the risk of more blood clots forming in blood vessels.

(a) (i) Describe how a blood clot forms in a blood vessel.

**(3)**

(ii) Explain how a blood clot in the coronary artery increases the risk of a heart attack.

**(3)**

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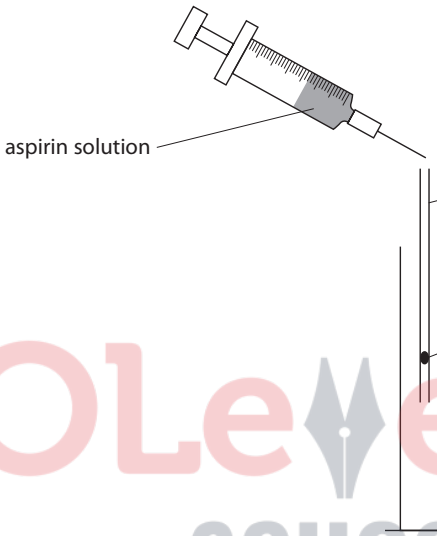
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## 3. Question 9 · part b.i · 1 mark

June 2019 Paper 2

Further practice · Attempt part b.i only.

(b) A student uses this apparatus to investigate the effect of aspirin on the rate of blood flow through blood vessels.




The student uses this method.

- dissolve one aspirin tablet in water
- use a syringe to pass the aspirin solution through a straw containing a blood clot
- record the total volume of aspirin solution in the measuring cylinder every five minutes

The student repeats the method with solutions formed from two aspirin tablets and then from three aspirin tablets.

(i) Give a control variable for this investigation.

**(1)**

.....

(ii) Describe a suitable control test for this investigation.

**(1)**

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.....

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## 3. Question 9 · part b.i · 1 mark · Working and answer

Working and answer

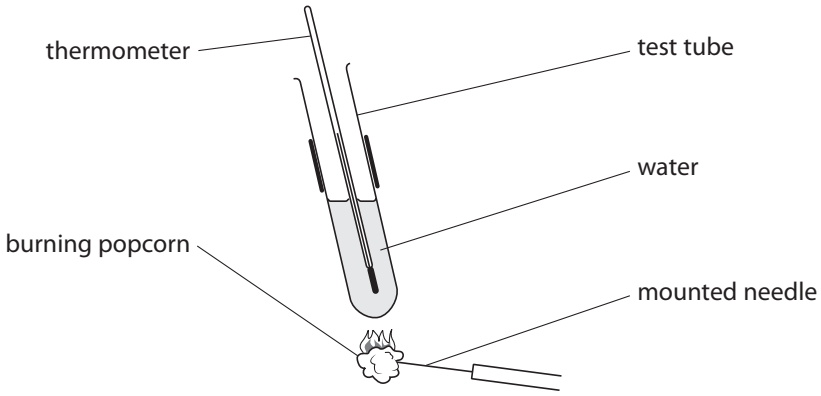
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# 4. Question 2 · part c · 4 marks

November 2020 Paper 1

Further practice · Attempt part c only.

**2** A student uses this apparatus to investigate the energy contained in popcorn.




This is the student's method.

- place 12 g of water in a test tube
- measure the initial temperature of the water
- place the popcorn on a mounted needle
- light the popcorn using a Bunsen flame
- place the burning popcorn under the test tube
- measure the maximum temperature of the water

(a) (i) State why the test tube should have thin glass walls.

**(1)**

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.....

(ii) Explain one safety precaution that the student should take during the investigation.

**(2)**

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## 4. Question 2 · part c · 4 marks

November 2020 Paper 1

Further practice · Attempt part c only.

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(b) The student uses this equation to calculate the energy released by the popcorn.

energy released = mass (g) × 4.2 × temperature rise (°C)

The initial temperature of the 12 g of water is 17 °C.

The maximum temperature of the water is 45 °C.

(i) Calculate the energy released when the student burns the popcorn.

Give your answer to 2 significant figures.

**(3)**

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energy released = ..... J

(ii) Explain why the calculated amount of energy released is not the same as the full amount of energy contained in the popcorn.

**(3)**

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## 4. Question 2 · part c · 4 marks

November 2020 Paper 1

Further practice · Attempt part c only.

(c) The student wants to compare the energy content of popcorn with the energy content of two other foods.

State four factors that the student should control to make sure the comparison is valid.

**(4)**

1 .....

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2 .....

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3 .....

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4 .....

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**(Total for Question 2 = 13 marks)**

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## 4. Question 2 · part c · 4 marks · Working and answer

Working and answer

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## 5. Question 7 · part b.i · 3 marks

November 2020 Paper 2R

Further practice · Attempt part b.i only.

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**7** Chymosin is an enzyme found in young humans, but not in adults. It converts soluble proteins in milk into solid proteins.

(a) (i) Suggest the advantages to a young human of having chymosin in their alimentary canal.

**(2)**

(ii) Explain why chymosin is only needed in young humans.

**(2)**

(iii) Describe a test to show that the solid formed by the action of chymosin on milk is a protein.

**(3)**

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# 5. Question 7 · part b.i · 3 marks

November 2020 Paper 2R

Further practice · Attempt part b.i only.

(b) A student investigates the effect of carbon dioxide on the activity of chymosin.

The student bubbles different volumes of carbon dioxide gas into five samples of milk.

He then adds chymosin to each sample and records the time taken for the milk protein to become solid.

The student repeats this method three times.

The table shows the student's results.

| Number of bubbles of $CO_2$ | Time taken for chymosin to turn milk protein solid, in seconds | | | | |
|---|---|---|---|---|---|
| | Test 1 | Test 2 | Test 3 | Test 4 | Mean |
| 100 | 253 | 257 | 250 | 260 | 255 |
| 150 | 238 | 232 | 241 | 229 | |
| 200 | 216 | 214 | 219 | 211 | 215 |
| 250 | 208 | 202 | 212 | 198 | 205 |
| 300 | 210 | 200 | 199 | 311 | 203 |

(i) State three factors that the student should control.

**(3)**

1 .....

.....

2 .....

.....

3 .....

.....

(ii) Calculate the missing mean (average) time taken for 150 bubbles.

**(2)**

mean time taken = .....

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20

P 6 2 0 5 4 A 0 2 0 2 4

## 5. Question 7 · part b.i · 3 marks · Working and answer

Working and answer

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# 6. Question 4 · part a.ii · 2 marks

November 2020 Paper 2

Further practice · Attempt part a.ii only.

**4** Aspirin is the active ingredient used in some tablets to relieve pain.

Aspirin becomes acidic when it dissolves in the body. The faster aspirin dissolves, the faster it can relieve pain.

A student investigates if it is better to mix sucrose or starch with the active ingredient to make an aspirin tablet.

This is the student's method.

- add distilled water to three beakers, A, B and C
- add 1 aspirin tablet to each beaker
- add 5 g of sucrose to beaker B
- add 5 g of starch to beaker C
- measure the pH of the contents of each beaker at the start of the investigation
- measure the pH of the contents of each beaker every minute for 5 minutes

The table shows the student's results.

| Beaker | pH of contents of beaker | | | | | |
|---|---|---|---|---|---|---|
| | at start | 1 minute | 2 minutes | 3 minutes | 4 minutes | 5 minutes |
| A – aspirin | 7.6 | 7.1 | 6.6 | 6.2 | 5.8 | 5.4 |
| B – aspirin and sucrose | 7.2 | 6.7 | 6.3 | 5.7 | 5.2 | 4.7 |
| C – aspirin and starch | 7.1 | 7.2 | 7.3 | 7.3 | 7.4 | 7.4 |

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# 6. Question 4 · part a.ii · 2 marks

November 2020 Paper 2

Further practice · Attempt part a.ii only.

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(a) (i) Plot a graph of the student's results.

Join the points with straight lines.

**(5)**



(ii) State two variables that should be controlled in this investigation.

**(2)**

1 .....

2 .....

(iii) Name the dependent variable in this investigation.

**(1)**

.....

## 6. Question 4 · part a.ii · 2 marks · Working and answer

Working and answer

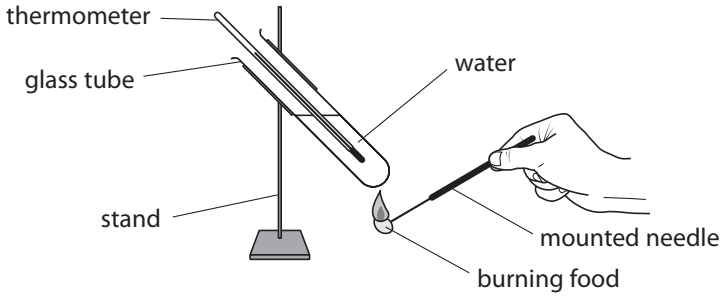
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# 7. Question 7 · part a.ii · 3 marks

June 2021 Paper 1

Further practice · Attempt part a.ii only.

**7** A student uses this apparatus to investigate the energy content of different foods.




This is the student's method.

- add water to the glass tube
- measure the temperature of the water
- place a piece of food onto the mounted needle
- set the food alight by placing it into a Bunsen flame
- place the burning food under the glass tube
- measure the temperature of the water when the food stops burning

Repeat the method using different foods.

(a) (i) State three safety precautions that the student should take.

**(3)**

1 .....

.....

2 .....

.....

3 .....

.....

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P 6 6 4 3 2 A 0 1 8 2 4

# 7. Question 7 · part a.ii · 3 marks

June 2021 Paper 1

Further practice · Attempt part a.ii only.

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(ii) State three factors that should be controlled during the investigation.

**(3)**

1 .....

.....

2 .....

.....

3 .....

.....

(iii) Explain why the energy content of a sample of food is likely to be higher than the energy of the same sample measured by this method.

**(5)**

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## 7. Question 7 · part a.ii · 3 marks · Working and answer

Working and answer

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# 8. Question 7 · part a.i · 3 marks

June 2021 Paper 2

Further practice · Attempt part a.i only.

**7** Garlic plants contain a substance called allicin. This substance is thought to have antibacterial properties.

A student investigates the antibacterial effect of garlic.

This is the student's method.

- label three test tubes A, B and C
- add 10 $cm^3$ of milk into each of the tubes
- add a sample of the bacterium *E. coli* into tubes B and C using aseptic techniques
- crush the garlic in water to form garlic juice
- add 1 $cm^3$ of garlic juice to tube C
- shake all three tubes and leave to incubate for two hours
- place a sample from tube A into the centre of Petri dish A, from tube B into the centre of Petri dish B and from tube C into the centre of Petri dish C
- incubate the three Petri dishes for five days
- each day measure the diameter of any bacterial growth that has occurred

The table shows the student's results.

| Petri dish | Diameter of bacterial growth in mm | | | | |
|---|---|---|---|---|---|
| | day 1 | day 2 | day 3 | day 4 | day 5 |
| A | 0 | 1.7 | 3.0 | 4.6 | 7.1 |
| B | 0 | 4.2 | 8.4 | 15.1 | 36.5 |
| C | 0 | 0 | 0 | 0 | 0 |

(a) (i) State three control variables in the student's investigation.

**(3)**

1 .....

.....

2 .....

.....

3 .....

.....

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## 8. Question 7 · part a.i · 3 marks

June 2021 Paper 2

Further practice · Attempt part a.i only.

(ii) State the dependent variable in the student's investigation.

**(1)**

.....

(b) (i) Calculate the mean daily increase in the diameter of the bacterial growth for dish A and dish B.

**(2)**

mean daily increase for dish A ..... mm/day

mean daily increase for dish B ..... mm/day

(ii) Explain the difference between the mean daily increases in the diameter of the bacterial growth for dish A and dish B.

**(4)**

.....

.....

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.....

(c) Explain what is meant by an aseptic techniques for the transfer of bacteria.

**(2)**

.....

.....

.....

.....

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## 8. Question 7 · part a.i · 3 marks · Working and answer

Working and answer

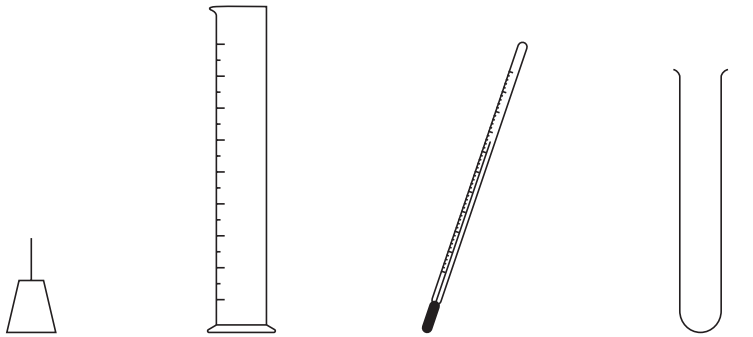
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# 9. Question 3 · part a.ii · 2 marks

June 2022 Paper 1

Further practice · Attempt part a.ii only.

**3** (a) The diagram shows some equipment that can be used to investigate the energy content of different foods.



mounted needle | measuring cylinder | thermometer | test tube

(i) Describe how the equipment can be used to investigate the energy content of different foods.

**(5)**

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# 9. Question 3 · part a.ii · 2 marks

June 2022 Paper 1

Further practice · Attempt part a.ii only.

(ii) Name two variables that need to be controlled in the investigation.

**(2)**

1 .....

.....

2 .....

.....

(iii) Give two safety precautions needed for this investigation.

**(2)**

1 .....

.....

2 .....

.....

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## 9. Question 3 · part a.ii · 2 marks · Working and answer

Working and answer

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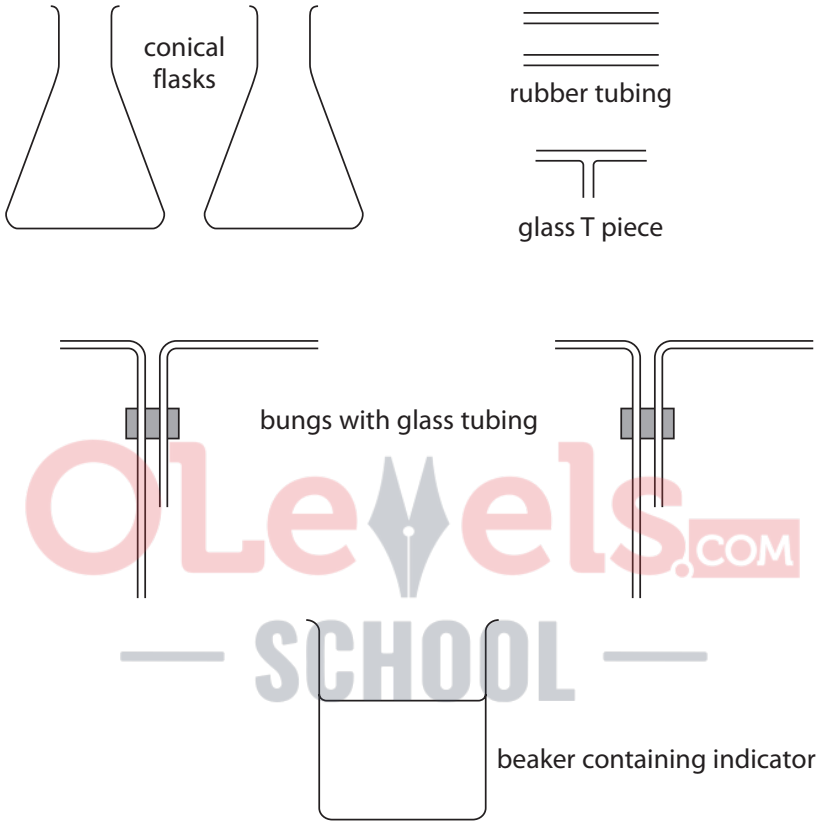
## 10. Question 1 · part e · 1 mark

June 2022 Paper 2R

Further practice · Attempt part e only.

**1** A student investigates whether there is more carbon dioxide in expired air than in inspired air.

This is the apparatus the student uses.




The apparatus is not drawn to scale.

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## 10. Question 1 · part e · 1 mark

June 2022 Paper 2R

Further practice · Attempt part e only.

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(a) Draw a diagram to show how the apparatus should be put together for the investigation.

**(3)**

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(b) Name an indicator that could be used in the investigation.

**(1)**

(c) Describe how the apparatus would be used to determine whether there is more carbon dioxide in expired air than in inspired air.

**(4)**

## 10. Question 1 · part e · 1 mark

June 2022 Paper 2R

Further practice · Attempt part e only.

(d) State one safety precaution.

**(1)**

.....

.....

(e) State one variable that should be controlled.

**(1)**

.....

.....

**(Total for Question 1 = 10 marks)**

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Working and answer

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# 11. Question 6 · part a.ii · 2 marks

January 2023 Paper 1

Further practice · Attempt part a.ii only.

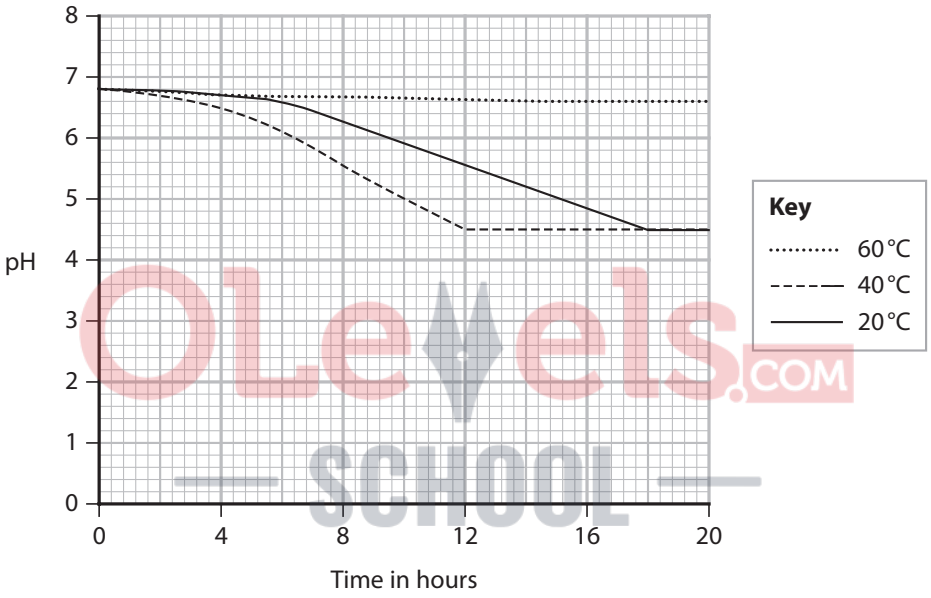
**6** Yoghurt can be produced by incubating milk with a small amount of live yoghurt.

The live yoghurt contains *Lactobacillus* bacteria.

Enzymes convert lactose in the milk into lactic acid causing the pH to decrease to around pH 4.4.

In an investigation, the change in pH as yoghurt is produced was measured during 20 hours at three different temperatures.

The graph shows the results.




(a) (i) Calculate the difference in the time taken to produce yoghurt at 40 °C instead of 20 °C.

**(2)**

time = ..... hours

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## 11. Question 6 · part a.ii · 2 marks

January 2023 Paper 1

Further practice · Attempt part a.ii only.

(ii) Give two variables that need to be controlled in this investigation.

**(2)**

1 .....

.....

2 .....

.....

(iii) Describe how this investigation could be improved to find the optimum temperature for the production of yoghurt.

**(2)**

.....

.....

.....

.....

.....

(iv) In some methods to produce yoghurt, the milk is boiled and then cooled before the bacteria are added.

Give the reason for boiling and the reason for cooling the milk.

**(2)**

boiling

.....

.....

cooling

.....

.....

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## 11. Question 6 · part a.ii · 2 marks · Working and answer

Working and answer

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# 12. Question 3 · part b.ii · 2 marks

January 2023 Paper 2

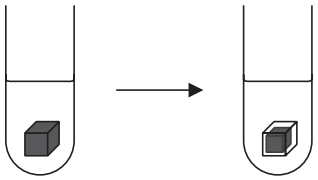
Further practice · Attempt part b.ii only.

**3** A student investigates diffusion using agar cubes and an acid.

The agar cube contains an alkali and a pH indicator.

The indicator is pink in alkali and colourless in acid.

Acid diffuses into the cube, causing the cube to change from pink to colourless from the edge.



agar cube when first placed in acid

agar cube starts to become colourless

(a) The cube the student placed in the acid had a side length of 2 mm.

(i) Calculate the surface area to volume ratio of the cube.

Your answer should be in the form n : 1

**(3)**

surface area to volume ratio = .....

(ii) The cube took 1 minute and 32 seconds to become colourless.

Which of these shows the time in seconds?

**(1)**

- ☒ **A** 62
- ☒ **B** 82
- ☒ **C** 92
- ☒ **D** 102

(iii) State what is meant by the term **diffusion**.

**(1)**

.....

.....

.....

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## 12. Question 3 · part b.ii · 2 marks

January 2023 Paper 2

Further practice · Attempt part b.ii only.

(iv) Explain the effect of increasing the surface area to volume ratio on the time taken for the cube to become colourless.

**(2)**

(b) A student wants to investigate the effect of temperature on the rate of diffusion into the agar cube.

(i) Describe a method for this investigation using the apparatus shown in the diagram.

**(2)**

(ii) Give two variables the student would need to control in their investigation.

**(2)**

1 .....

2 .....

**(Total for Question 3 = 11 marks)**

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## 12. Question 3 · part b.ii · 2 marks · Working and answer

Working and answer

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# 13. Question 2 · part b.iv · 1 mark

June 2023 Paper 2R

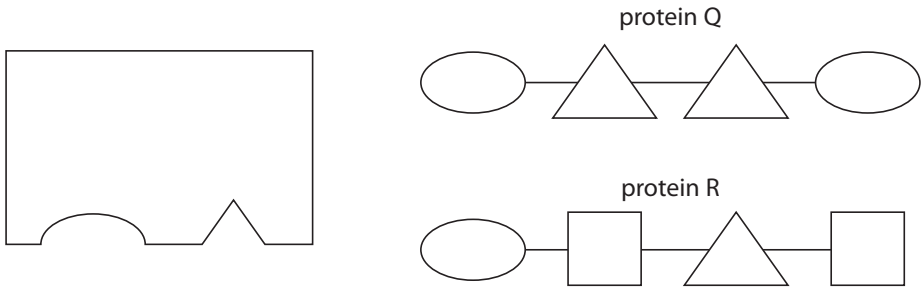
Further practice · Attempt part b.iv only.

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**2** (a) The diagram represents an enzyme, known as enzyme P, and two named substrates.




**Enzyme P** **Substrates**

(i) Use the diagram and words from the box to complete the passage about enzyme P.

**(3)**

| amino acids | amylases | fatty acids | proteases |
|---|---|---|---|
| protein Q | protein R | starch | sugar |

Enzyme P belongs to a group of enzymes called ..... .

This enzyme breaks down ..... in food

to ..... .

(ii) Give one location where enzyme P is found in the digestive system.

**(1)**

.....

(iii) Name the molecule that codes for the production of enzyme P.

**(1)**

.....

# 13. Question 2 · part b.iv · 1 mark

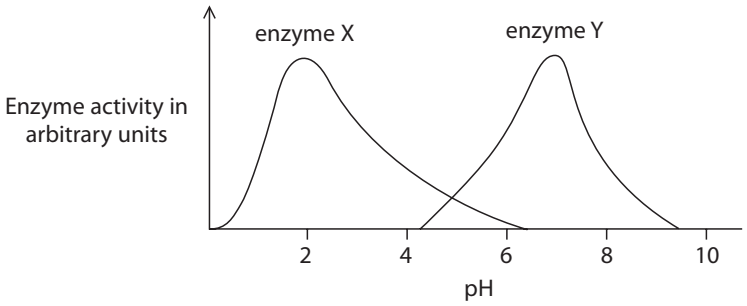
June 2023 Paper 2R

Further practice · Attempt part b.iv only.

(b) A student investigates the effect of pH on the activity of two enzymes, X and Y.

The student measures the activity of each enzyme once, using different pH values.

The graph shows the results of their investigation.




(i) Describe how the activity of enzyme X changes with pH.

**(3)**

(ii) Explain why no activity is shown for enzyme Y at pH2.

**(2)**

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P 7 2 5 8 8 A 0 6 2 4

## 13. Question 2 · part b.iv · 1 mark

June 2023 Paper 2R

Further practice · Attempt part b.iv only.

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(iii) Describe how the student could make the results of this investigation more reliable.

**(2)**

.....

.....

.....

.....

(iv) Name one control variable in the student's investigation.

**(1)**

.....

**(Total for Question 2 = 13 marks)**

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Working and answer

# 14. Question 7 · part a.i · 3 marks

November 2023 Paper 2

Further practice · Attempt part a.i only.

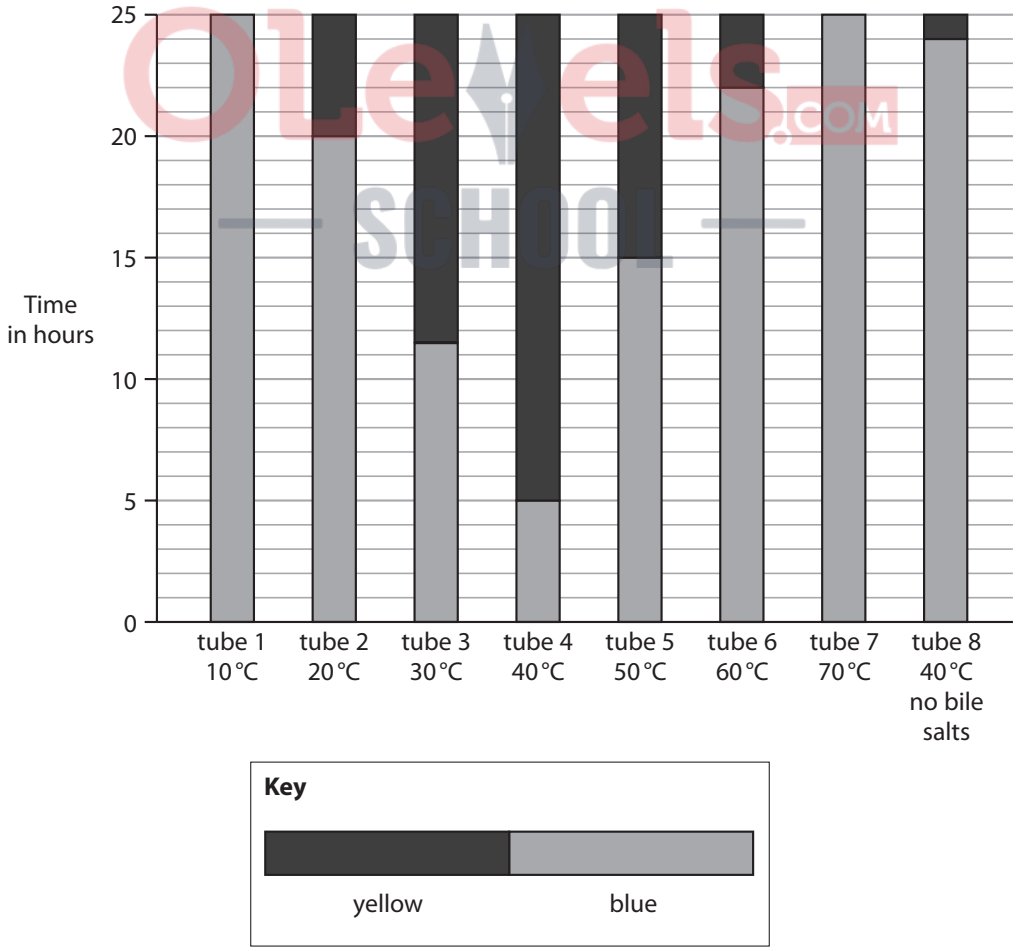
**7** A student investigates the digestion of olive oil (lipid) using the enzyme lipase, bile salts and a blue indicator.

The indicator changes from blue to yellow when the pH is below 5.

This is the student's method.

- mix together olive oil, lipase solution, bile salts and one drop of indicator
- put this mixture into a test tube
- repeat the process for six other test tubes
- keep the tubes at different temperatures
- keep another test tube (test tube 8) containing just olive oil, indicator and lipase at 40 °C
- record the colours of the mixture of each tube every hour for 25 hours

The chart shows the student's results.




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# 14. Question 7 · part a.i · 3 marks

November 2023 Paper 2

Further practice · Attempt part a.i only.

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(a) (i) State three factors that should be kept constant in the student's investigation.

**(3)**

1 .....

2 .....

3 .....

(ii) State the optimum temperature for the action of lipase.

**(1)**

.....

(iii) Discuss why this temperature is only an approximate optimum temperature.

**(3)**

.....

.....

.....

.....

.....

.....

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(b) (i) Determine how much more time the indicator takes to change colour in tube 8 than in tube 3.

**(2)**

time = ..... hours

(ii) Explain why the pH decreases during the 25 hours.

**(3)**

.....

.....

.....

.....

.....

.....

## 14. Question 7 · part a.i · 3 marks · Working and answer

Working and answer

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# 15. Question 2 · part d · 2 marks

November 2024 Paper 1

Further practice · Attempt part d only.

**2** A student investigates the activity of an enzyme on fat found in milk.
This is the student's method.

- pour milk and enzyme solution into a test tube
- determine the pH of the solution every minute for 10 minutes

The table shows the student's results.

| Time in minutes | pH of solution |
|---|---|
| 1 | 6.8 |
| 2 | 6.4 |
| 3 | 6.0 |
| 4 | 5.8 |
| 5 | 5.6 |
| 6 | 5.2 |
| 7 | 5.0 |
| 8 | 4.8 |
| 9 | 4.6 |
| 10 | 4.6 |

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## 15. Question 2 · part d · 2 marks

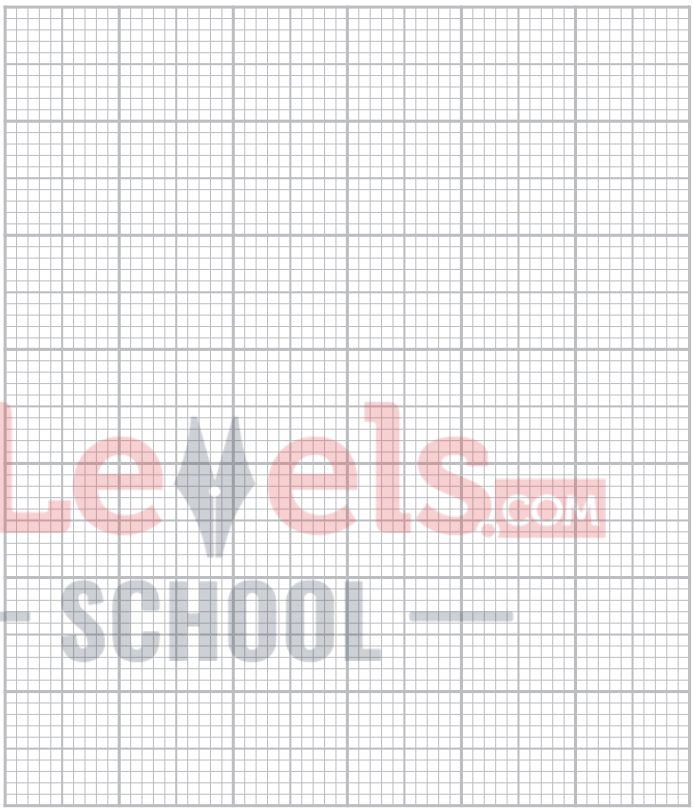
November 2024 Paper 1

Further practice · Attempt part d only.

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(a) (i) Plot a graph of the student's results.

**(3)**



(ii) Draw a line of best fit.

**(1)**

(iii) Give the trend shown by the graph.

**(1)**

.....

.....

## 15. Question 2 · part d · 2 marks

November 2024 Paper 1

Further practice · Attempt part d only.

(b) Explain the difference in the pH at 1 minute and the pH at 10 minutes.

**(3)**

(c) Name the independent variable in this investigation.

**(1)**

(d) Give two variables that the student should control in this investigation.

**(2)**

1 .....

2 .....

(e) State how the student could improve the reliability of their results.

**(1)**

**(Total for Question 2 = 12 marks)**

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## 15. Question 2 · part d · 2 marks · Working and answer

Working and answer

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# 16. Question 4 · part b.iii · 3 marks

June 2025 Paper 1R

Further practice · Attempt part b.iii only.

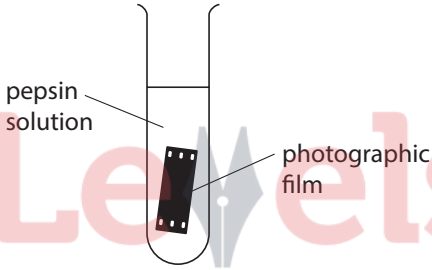
**4** Photographic film is made of clear plastic, coated in a protein. The protein contains silver compounds which give the film its black colour.

Pepsin is an enzyme that digests protein.

A student investigates how the concentration of pepsin affects the time taken to digest the protein in a piece of photographic film.

This is the student's method.

- place a piece of photographic film into a test tube
- add a 1% pepsin solution
- measure the time for the film to become clear
- repeat the investigation using other concentrations of pepsin




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The table shows the student's results.

| Percentage pepsin concentration (%) | Time taken for film to become clear in seconds |
|---|---|
| 1 | 112 |
| 2 | 102 |
| 3 | 93 |
| 4 | 84 |
| 5 | 84 |
| 6 | 84 |

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# 16. Question 4 · part b.iii · 3 marks

June 2025 Paper 1R

Further practice · Attempt part b.iii only.

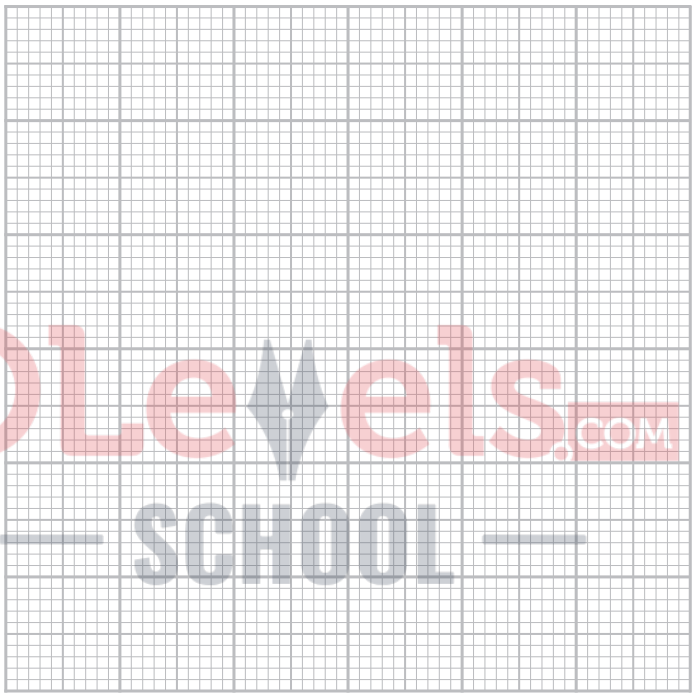
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(a) (i) Plot a graph of the results on the grid.

**(4)**

(ii) Draw a curve of best fit.

**(1)**



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(b) (i) Explain why the photographic film becomes clear.

**(3)**

## 16. Question 4 · part b.iii · 3 marks

June 2025 Paper 1R

Further practice · Attempt part b.iii only.

(ii) Explain which would be the best pH to carry out this investigation.

**(2)**

(iii) State three variables that should be controlled during this investigation other than pH.

**(3)**

1 .....

2 .....

3 .....

(c) Suggest why the time taken for the film to become clear is the same for 4%, 5% and 6% concentrations of pepsin.

**(2)**

**(Total for Question 4 = 15 marks)**

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## 16. Question 4 · part b.iii · 3 marks · Working and answer

Working and answer

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# 17. Question 4 · part c.ii · 2 marks

June 2025 Paper 2

Further practice · Attempt part c.ii only.

**4** A student investigates the effect of different concentrations of sugar solution on the mass of cylinders made from potato.

This is the student's method.

- push a cork borer into different parts of a potato
- remove five cylinders of potato
- wrap each cylinder in a piece of filter paper
- remove from the filter paper and find the mass of each cylinder
- place one cylinder into distilled water
- place each of the other cylinders into different concentrations of sugar solution
- remove the cylinders after two hours and wrap in filter paper
- remove from the filter paper and find the new mass

These are the student's results.

| | | |
|---|---|---|
| 1.00 au = −7 % | 1.50 au = −10 % | 0.00 au (distilled water) = +8 % |
| | 0.25 au = +4 % | 0.50 au = −1 % |

The student's results are given as the percentage change in mass of the cylinder and the concentration of sugar is given in arbitrary units (au).

(a) (i) Draw a table to display the results.

**(4)**

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P 7 8 7 6 9 A 0 8 2 0

# 17. Question 4 · part c.ii · 2 marks

June 2025 Paper 2

Further practice · Attempt part c.ii only.

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(ii) Determine the median value for sugar concentration.

**(1)**

median value = .....

(iii) The original mass of the potato cylinder placed into the sugar solution, of concentration 0.25 au, was 7.0 g.

Calculate the mass of the cylinder after two hours.

Give your answer to two significant figures.

**(3)**

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mass of cylinder = ..... g

(b) Use the student's results to estimate the concentration of sugar solution that would produce no change in the mass of the potato cylinder.

You should include details of how you obtained your answer.

**(4)**

concentration of sugar solution = ..... au

## 17. Question 4 · part c.ii · 2 marks

June 2025 Paper 2

Further practice · Attempt part c.ii only.

(c) (i) Explain why filter paper was wrapped around the potato cylinders at the beginning and end of the investigation.

**(2)**

(ii) The student kept the potato cylinders in the liquid for two hours.

State two other factors that should be kept constant during this investigation.

**(2)**

1 .....

2 .....

**(Total for Question 4 = 16 marks)**

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Working and answer

## 18. Question 2 · part c · 1 mark

January 2020 Paper 2

Further practice · Attempt part c only.

**2** A student records her pulse rate in beats per minute (bpm) after different activities.

These are the student's results.

- at rest 78 bpm
- walking 95 bpm
- jogging 138 bpm
- running 146 bpm

(a) Draw a table to show the student's results.

**(3)**

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## 18. Question 2 · part c · 1 mark

January 2020 Paper 2

Further practice · Attempt part c only.

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(b) Describe a method that the student could use to obtain these results.

**(4)**

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(c) State one factor that the student should control during the investigation.

**(1)**

(d) Name a body process, other than pulse rate, that increases during exercise.

**(1)**

**(Total for Question 2 = 9 marks)**

## 18. Question 2 · part c · 1 mark · Working and answer

Working and answer

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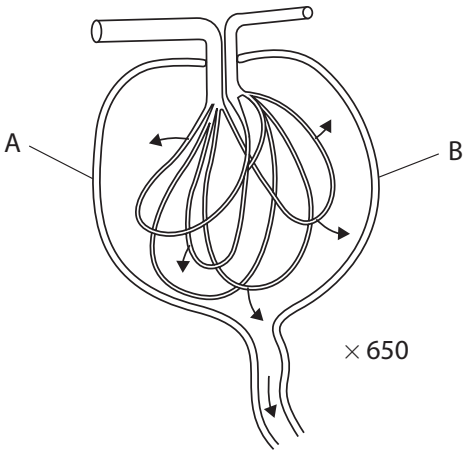
## 19. Question 1 · part b.i · 2 marks

January 2022 Paper 2

Further practice · Attempt part b.i only.

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**1** (a) The diagram shows the Bowman's capsule and glomerulus of a kidney tubule (nephron).




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(i) Complete the diagram to show the missing parts of the kidney tubule.

**(3)**

(ii) Calculate the actual width of the Bowman's capsule, in mm, from the point labelled A to the point labelled B.

**(3)**

width of Bowman's capsule = ..... mm



P 6 7 0 6 1 A 0 3 2 4

# 19. Question 1 · part b.i · 2 marks

January 2022 Paper 2

Further practice · Attempt part b.i only.

(iii) Name two components of the blood that would not pass into the Bowman's capsule.

**(2)**

1 .....

2 .....

(iv) Name one component of the glomerular filtrate that is completely reabsorbed into the blood from the tubule.

**(1)**

.....

(b) A scientist uses this method to investigate the effect of room temperature on urine production.

On day 1, a person drinks 1 $dm^3$ of water. Urine samples from the person are collected at intervals during a time period of four hours.

The room temperature is 20 °C.

This is then repeated on day 2, when the room temperature is 30 °C.

The table shows the results.

| Time after drinking in hours | Volume of urine in $cm^3$ | |
|---|---|---|
| | day 1 (20 °C) | day 2 (30 °C) |
| 0.5 | 70 | 50 |
| 1.5 | 175 | 70 |
| 2.0 | 300 | 200 |
| 2.5 | 250 | 190 |
| 3.0 | 210 | 150 |
| 4.0 | 100 | 50 |

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## 19. Question 1 · part b.i · 2 marks

January 2022 Paper 2

Further practice · Attempt part b.i only.

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(i) State two factors that should be controlled in the investigation.

**(2)**

1 .....

.....

2 .....

.....

(ii) The mean rate of urine production on day 1 is 276 $cm^3$ per hour.

Calculate the mean rate of urine production on day 2 during the 4 hour time period.

**(2)**

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mean rate of urine production on day 2 = ..... $cm^3$ per hour

(iii) Explain the difference between the mean rate on day 1 and the mean rate on day 2.

**(3)**

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**(Total for Question 1 = 16 marks)**

## 19. Question 1 · part b.i · 2 marks · Working and answer

Working and answer

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## 20. Question 5 · part a.ii · 3 marks

June 2023 Paper 1R

Further practice · Attempt part a.ii only.

**5** A student uses this method to investigate the effect of exercise on the production of sweat.

- place a person in a room which has a temperature of 20 °C
- ask the person to remain inactive for 5 minutes
- measure the amount of sweat produced by a 25 $cm^2$ area of skin
- ask the person to exercise and measure the amount of sweat produced
- ask the person to rest for 10 minutes

Repeat the method at room temperatures of 22 °C, 24 °C, 26 °C, 28 °C and 30 °C

(a) (i) Explain why the person rests for 10 minutes before the method is repeated.

**(2)**

(ii) State three variables that should be controlled.

**(3)**

1 .....

2 .....

3 .....

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## 20. Question 5 · part a.ii · 3 marks

June 2023 Paper 1R

Further practice · Attempt part a.ii only.

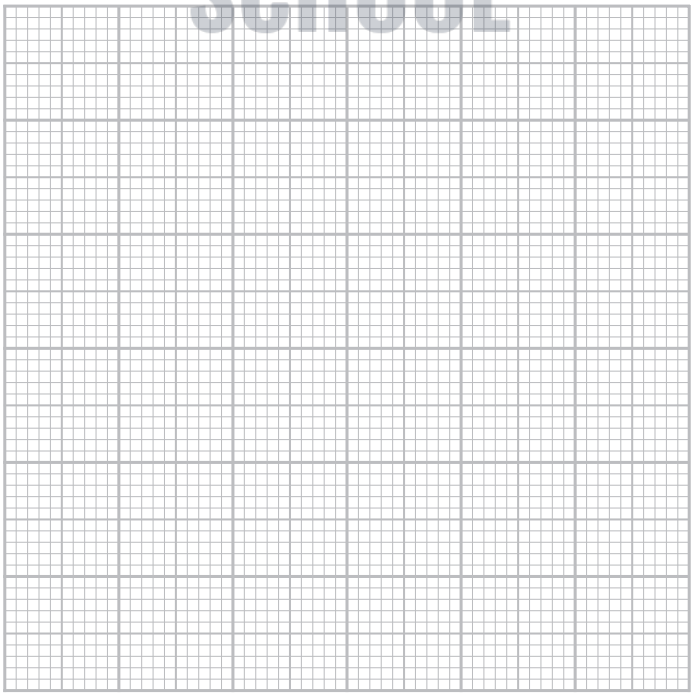
(b) The table shows the student's results.

| Room temperature in °C | Amount of sweat produced by 25 $cm^2$ area of skin in arbitrary units | |
|---|---|---|
| | While inactive | During exercise |
| 20 | 1 | 2 |
| 22 | 2 | 6 |
| 24 | 2 | 8 |
| 26 | 2 | 11 |
| 28 | 3 | 14 |
| 30 | 4 | 22 |

(i) Plot line graphs of the student's results.

Join the points with straight lines.

**(6)**



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P 7 2 5 8 6 A 0 1 2 2 0

## 20. Question 5 · part a.ii · 3 marks · Working and answer

Working and answer

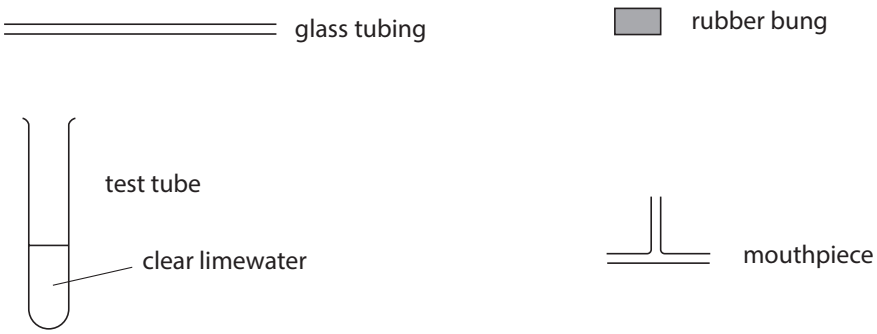
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# 21. Question 4 · part b.i · 1 mark

June 2024 Paper 1

Further practice · Attempt part b.i only.

**4** (a) The diagram shows some scientific equipment.




(i) Draw a diagram to show how this equipment is set up to investigate the difference between the carbon dioxide concentration of inspired air and of expired air.

**(3)**

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# 21. Question 4 · part b.i · 1 mark

June 2024 Paper 1

Further practice · Attempt part b.i only.

(ii) State one safety precaution in this investigation.

**(1)**

(b) A student investigates their breathing rate at rest, after gentle exercise and again after vigorous exercise.

The student repeats each exercise three times.

The student's mean results are shown in the table.

| Mean breathing rate in breaths per minute | | |
|---|---|---|
| **at rest** | **after gentle exercise** | **after vigorous exercise** |
| 14 | 19 | 50 |

(i) Give one control variable in this investigation.

**(1)**

(ii) Explain how the breathing rate after vigorous exercise benefits working muscles.

**(4)**

**(Total for Question 4 = 9 marks)**

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## 21. Question 4 · part b.i · 1 mark · Working and answer

Working and answer

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# Practice workbook · Describe a control or explain why a step is in a method

Control tubes, discs or beakers without the active ingredient, and justified processing steps such as blotting, rinsing or using percentage change.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-PRAC-EXPLAIN-CONTROL-OR-METHOD-STEP

# 1. Question 8 · part b.iii · 3 marks

June 2023 Paper 1R

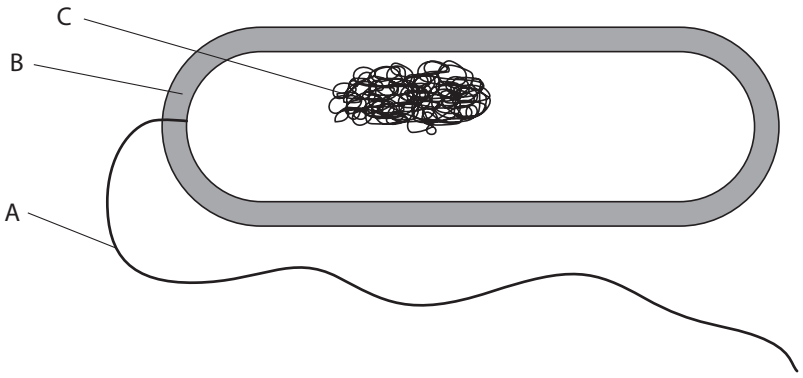
Worked-example retry · Attempt part b.iii only.

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**8** The diagram shows a bacterial cell.




(a) (i) Name the parts labelled A, B and C.

**(3)**

A .....

B .....

C .....

(ii) Some types of bacteria cause disease.

Give the name of organisms that cause disease.

**(1)**

.....

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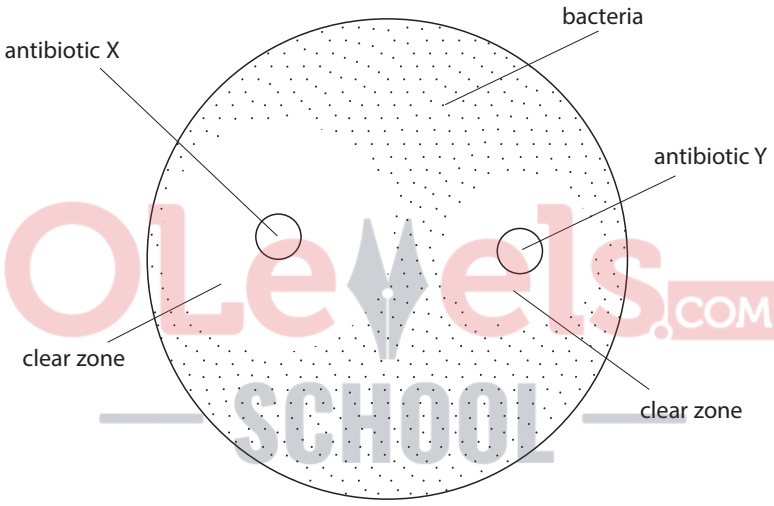
# 1. Question 8 · part b.iii · 3 marks

June 2023 Paper 1R

Worked-example retry · Attempt part b.iii only.

(b) A student uses this method to investigate whether antibiotic X is more effective than antibiotic Y.

- grow bacteria on nutrient agar in a Petri dish
- soak one disc of filter paper in antibiotic X and another disc in antibiotic Y
- place each disc on the agar in the Petri dish
- seal the Petri dish and leave for three days




(i) Describe what is meant by the term **antibiotic**.

**(2)**

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P 7 2 5 8 6 A 0 1 8 2 0

# 1. Question 8 · part b.iii · 3 marks

June 2023 Paper 1R

Worked-example retry · Attempt part b.iii only.

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(ii) Explain which antibiotic, X or Y, is more effective against the bacteria in the Petri dish.

**(3)**

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(iii) Explain a suitable control that could be used in this investigation.

**(3)**

**(Total for Question 8 = 12 marks)**

## 1. Question 8 · part b.iii · 3 marks · Working and answer

Working and answer

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## 2. Question 9 · part b.ii · 1 mark

June 2019 Paper 2

Further practice · Attempt part b.ii only.

**9** Aspirin is a painkiller, but can also be prescribed to people who are at risk of having a stroke or a heart attack.

Aspirin dissolves blood clots and is also an enzyme inhibitor, which reduces the risk of more blood clots forming in blood vessels.

(a) (i) Describe how a blood clot forms in a blood vessel.

**(3)**

(ii) Explain how a blood clot in the coronary artery increases the risk of a heart attack.

**(3)**

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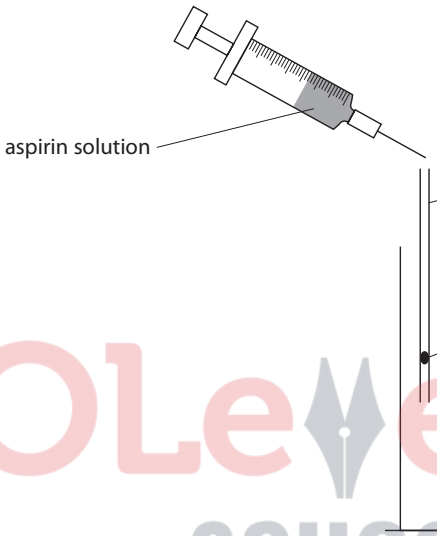
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## 2. Question 9 · part b.ii · 1 mark

June 2019 Paper 2

Further practice · Attempt part b.ii only.

(b) A student uses this apparatus to investigate the effect of aspirin on the rate of blood flow through blood vessels.




The student uses this method.

- dissolve one aspirin tablet in water
- use a syringe to pass the aspirin solution through a straw containing a blood clot
- record the total volume of aspirin solution in the measuring cylinder every five minutes

The student repeats the method with solutions formed from two aspirin tablets and then from three aspirin tablets.

(i) Give a control variable for this investigation.

**(1)**

(ii) Describe a suitable control test for this investigation.

**(1)**

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P 5 8 5 6 6 A 0 2 4 2 8

## 2. Question 9 · part b.ii · 1 mark

June 2019 Paper 2

Further practice · Attempt part b.ii only.

The table shows the student's results.

| Time in minutes | Total volume of aspirin solution in the measuring cylinder in $cm^3$ | | |
|---|---|---|---|
| | 1 aspirin tablet | 2 aspirin tablets | 3 aspirin tablets |
| 5 | 0 | 0 | 0 |
| 10 | 0 | 0 | 0 |
| 15 | 0 | 0 | 1 |
| 20 | 0 | 1 | 2 |
| 25 | 1 | 1 | 3 |
| 30 | 1 | 2 | 4 |
| 35 | 2 | 4 | 6 |
| 40 | 5 | 7 | 9 |
| 45 | 10 | 12 | 16 |
| 50 | 17 | 19 | 22 |
| 55 | 25 | 27 | 31 |
| 60 | 35 | 37 | 39 |

(iii) Explain the pattern shown by the results.

**(3)**

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## 2. Question 9 · part b.ii · 1 mark · Working and answer

Working and answer

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# 3. Question 2 · part b · 2 marks

November 2020 Paper 2

Further practice · Attempt part b only.

**2** A student investigates the effect of bile salts on the digestion of lipids. The student uses milk as a source of lipids.

The student sets up three test tubes.

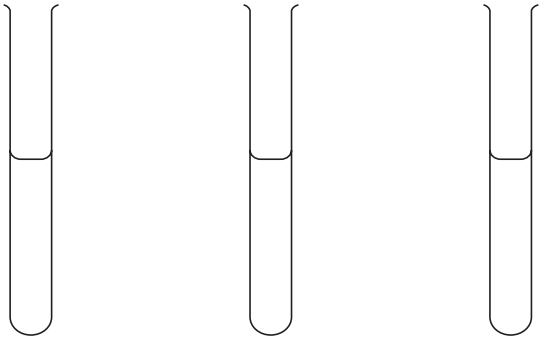



Table 1 lists the contents of each test tube.

| Test tube A | Test tube B | Test tube C |
|---|---|---|
| 5 $cm^3$ of milk | 5 $cm^3$ of milk | 5 $cm^3$ of milk |
| 2 $cm^3$ of sodium hydrogencarbonate | 2 $cm^3$ of sodium hydrogencarbonate | 2 $cm^3$ of sodium hydrogencarbonate |
| 6 drops of phenolphthalein | 6 drops of phenolphthalein | 6 drops of phenolphthalein |
| 1 $cm^3$ enzyme | 1 $cm^3$ enzyme | 1 $cm^3$ boiled enzyme |
| distilled water | bile salts | bile salts |

**Table 1**

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## 3. Question 2 · part b · 2 marks

November 2020 Paper 2

Further practice · Attempt part b only.

The student records the colour of the contents of each tube at the start of the investigation, and at 5 minute intervals for 15 minutes.

Table 2 shows the student's results.

| | Colour of contents | | |
|---|---|---|---|
| | **Tube A** | **Tube B** | **Tube C** |
| at start | pink | pink | pink |
| after 5 minutes | pink | colourless | pink |
| after 10 minutes | pink | colourless | pink |
| after 15 minutes | colourless | colourless | pink |

**Table 2**

Phenolphthalein is pink in solutions above pH 10 and colourless in solutions below pH 8.

(a) (i) Which enzyme is used in this investigation? **(1)**

- ☒ **A** amylase
- ☒ **B** carbohydrase
- ☒ **C** lipase
- ☒ **D** protease

(ii) Which chemical elements are found in lipids? **(1)**

- ☒ **A** carbon, hydrogen and oxygen
- ☒ **B** carbon, hydrogen, oxygen and nitrogen
- ☒ **C** carbon, hydrogen, oxygen and sulfur
- ☒ **D** carbon, hydrogen, oxygen, nitrogen and sulfur

(iii) Where in the body are bile salts produced? **(1)**

- ☒ **A** brain
- ☒ **B** duodenum
- ☒ **C** liver
- ☒ **D** pancreas

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## 3. Question 2 · part b · 2 marks

November 2020 Paper 2

Further practice · Attempt part b only.

(b) Explain the purpose of tube C.

**(2)**

(c) Suggest one reason for the addition of the sodium hydrogencarbonate.

**(2)**

(d) Describe the effect of bile salts on lipid digestion in this investigation.

**(3)**

**(Total for Question 2 = 10 marks)**

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## 3. Question 2 · part b · 2 marks · Working and answer

Working and answer

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# 4. Question 1 · part b.ii · 1 mark

November 2021 Paper 1

Further practice · Attempt part b.ii only.

**1** The enzyme catalase is found in the liver.

Catalase acts as a catalyst when added to hydrogen peroxide solution, causing the solution to break down into water and oxygen.

A student investigates the increase in the rate of breakdown of hydrogen peroxide into water and oxygen using catalase.

This is the student's method.

- put 100 g of liver into 100 $cm^3$ of water, and mix into a smooth paste
- put 10 $cm^3$ of the paste into a beaker at room temperature
- add one drop of hydrogen peroxide to the paste
- assess how many oxygen bubbles are produced on a scale of 0 to 5, with 5 representing most bubbles and 0 representing no bubbles

The student repeats the investigation using different conditions.

The conditions he uses are:

- liver paste heated to 70 °C
- liver paste that has been kept at 3 °C for an hour
- acidified liver paste
- liver paste with an alkali added
- liver paste with no hydrogen peroxide added

These are the student's results.

*room temperature = 5  heated to 70 °C = 0*

*kept at 3 °C = 1  acid added = 2  alkali added = 3*

*no hydrogen peroxide added = 0*

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# 4. Question 1 · part b.ii · 1 mark

November 2021 Paper 1

Further practice · Attempt part b.ii only.

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(a) (i) Give the student's results in the form of a suitable table.

**(3)**

(ii) State why this investigation is not reliable.

**(1)**

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(iii) Give a reason why the results of this investigation are not accurate.

**(1)**

(iv) Describe how the investigation could be improved so that the conclusions could be more accurate and reliable.

**(3)**

# 4. Question 1 · part b.ii · 1 mark

November 2021 Paper 1

Further practice · Attempt part b.ii only.

(b) (i) State the condition where the enzyme was most active.

**(1)**

(ii) State the reason for including the beaker with no hydrogen peroxide added.

**(1)**

(iii) Explain the result obtained when the conditions were acidic.

**(4)**

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**(Total for Question 1 = 14 marks)**

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Working and answer

## 5. Question 2 · part a.i · 2 marks

June 2022 Paper 2R

Further practice · Attempt part a.i only.

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**2** A student investigates the effect of a competitive inhibitor on the action of the enzyme phosphatase.

This is the reaction catalysed by the enzyme.

phenolphthalein phosphate ⟶ phenolphthalein + phosphate

Phenolphthalein turns pink when an alkali is added.

This colour can be measured using a colorimeter. The more intense the colour, the higher the reading on the colorimeter.

This is the student's method.

- prepare six different concentrations of phenolphthalein phosphate solution
- place 5.0 $cm^3$ of each solution into a different test tube
- add 1.0 $cm^3$ of a competitive inhibitor and 1.0 $cm^3$ of the enzyme phosphatase into each test tube
- wait 30 minutes, then add 1.0 $cm^3$ of an alkali to each tube
- measure the intensity of the pink colour using a colorimeter

(a) (i) Give a suitable control for this investigation. **(2)**

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(ii) State three variables that should be kept constant in this investigation. **(3)**

1 .....

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2 .....

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3 .....

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## 5. Question 2 · part a.i · 2 marks · Working and answer

Working and answer

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# 6. Question 6 · part d · 1 mark

June 2022 Paper 2

Further practice · Attempt part d only.

**6** A student uses this method to investigate the effect of different antibacterial cleaners on the growth of bacteria.

- add bacterial culture to sterile agar in a Petri dish using an inoculating loop
- using a pipette, place two drops of an antibacterial cleaner onto a filter paper disc
- repeat using two other filter paper discs, each one soaked with a different antibacterial cleaner
- soak one filter paper disc in distilled water
- place the four filter paper discs onto the sterile agar
- place a lid on the Petri dish
- place the dish into an incubator at 25 °C for 5 days

(a) State the independent variable in this investigation.

**(1)**

.....

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(b) Give two improvements to this method that will reduce the risk of introducing unwanted microorganisms into the Petri dish.

**(2)**

1 .....

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2 .....

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## 6. Question 6 · part d · 1 mark

June 2022 Paper 2

Further practice · Attempt part d only.

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(c) Explain why it is important that the Petri dish is incubated at a temperature no higher than 25 °C.

**(2)**

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(d) Give a reason why the filter paper disc soaked in distilled water is included in the investigation.

**(1)**

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**(Total for Question 6 = 6 marks)**

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Working and answer

# 7. Question 7 · part d · 3 marks

November 2024 Paper 2

Further practice · Attempt part d only.

**7** Six students investigate the activity of an enzyme found in the alimentary canal.

This enzyme digests a cloudy suspension into a clear solution.

The results of the investigation are shown in the table.

| Student | Volume of enzyme solution in $cm^3$ | Temperature in °C | pH | Volume of cloudy suspension in $cm^3$ | Time tubes left in minutes | Appearance at end of investigation |
|---|---|---|---|---|---|---|
| 1 | 5 | 37 | 2 | 10 | 9 | cloudy |
| 2 | 5 | 37 | 2 | 10 | 20 | clear |
| 3 | 5 | 37 | 4 | 10 | 35 | clear |
| 4 | 5 | 37 | 4 | 10 | 20 | cloudy |
| 5 | 5 | 37 | 6 | 10 | 8 | cloudy |
| 6 | 5 | 37 | 6 | 10 | 40 | cloudy |

(a) Explain why the investigation was carried out at 37 °C.

**(2)**

(b) Explain which property of the enzyme the students are investigating.

**(2)**

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# 7. Question 7 · part d · 3 marks

November 2024 Paper 2

Further practice · Attempt part d only.

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(c) Comment on the results of the investigation.

In your answer include ideas of validity and accuracy.

**(4)**

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(d) Describe a control experiment that the students could have carried out.

**(3)**

# 7. Question 7 · part d · 3 marks

November 2024 Paper 2

Further practice · Attempt part d only.

(e) State a possible conclusion that the students could make from the results of this investigation.

**(2)**

(f) Explain where this enzyme is likely to be found in the alimentary canal.

**(2)**

**(Total for Question 7 = 15 marks)**

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## 7. Question 7 · part d · 3 marks · Working and answer

Working and answer

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# 8. Question 5 · part c.ii · 2 marks

June 2019 Paper 2R

Further practice · Attempt part c.ii only.

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**5** Substances move in and out of cells by diffusion, osmosis and active transport.

(a) Describe the process of active transport.

**(2)**

(b) Water moves in and out of potato cells by osmosis.

Describe a method to determine the concentration of the contents of potato cells.

A large potato is available and also sucrose solutions of various concentrations.

**(5)**

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# 8. Question 5 · part c.ii · 2 marks

June 2019 Paper 2R

Further practice · Attempt part c.ii only.

(c) The table shows results obtained from an investigation to determine the concentration of the contents of potato cells.

| | Concentration of sucrose solution in mol per dm³ | | | | |
|---|---|---|---|---|---|
| | **0.2** | **0.4** | **0.6** | **0.8** | **1.0** |
| Initial mass in g | 8.4 | 7.9 | 8.7 | 8.3 | 8.4 |
| Mass after soaking in g | 8.7 | 8.0 | 8.2 | 7.5 | 7.3 |
| Change in mass in g | + 0.3 | + 0.1 | – 0.5 | – 0.8 | – 1.1 |
| Percentage change in mass | + 3.6 | | – 5.8 | – 9.6 | – 13.1 |

(i) Calculate the percentage change in mass for 0.4 mol per dm³.

**(2)**

.....%

(ii) Explain why it is necessary to calculate the percentage change in mass rather than using the change in mass in g.

**(2)**

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# 8. Question 5 · part c.ii · 2 marks

June 2019 Paper 2R

Further practice · Attempt part c.ii only.

(iii) Explain how the results in the table could be used to determine the concentration of the contents of potato cells.

**(4)**

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**(Total for Question 5 = 15 marks)**

Working and answer

# 9. Question 4 · part c.i · 2 marks

June 2025 Paper 2

Further practice · Attempt part c.i only.

**4** A student investigates the effect of different concentrations of sugar solution on the mass of cylinders made from potato.

This is the student's method.

- push a cork borer into different parts of a potato
- remove five cylinders of potato
- wrap each cylinder in a piece of filter paper
- remove from the filter paper and find the mass of each cylinder
- place one cylinder into distilled water
- place each of the other cylinders into different concentrations of sugar solution
- remove the cylinders after two hours and wrap in filter paper
- remove from the filter paper and find the new mass

These are the student's results.

| | | |
|---|---|---|
| 1.00 au = −7 % | 1.50 au = −10 % | 0.00 au (distilled water) = +8 % |
| | 0.25 au = +4 % | 0.50 au = −1 % |

The student's results are given as the percentage change in mass of the cylinder and the concentration of sugar is given in arbitrary units (au).

(a) (i) Draw a table to display the results.

**(4)**

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P 7 8 7 6 9 A 0 8 2 0

# 9. Question 4 · part c.i · 2 marks

June 2025 Paper 2

Further practice · Attempt part c.i only.

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(ii) Determine the median value for sugar concentration.

**(1)**

median value = .....

(iii) The original mass of the potato cylinder placed into the sugar solution, of concentration 0.25 au, was 7.0 g.

Calculate the mass of the cylinder after two hours.

Give your answer to two significant figures.

**(3)**

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mass of cylinder = ..... g

(b) Use the student's results to estimate the concentration of sugar solution that would produce no change in the mass of the potato cylinder.

You should include details of how you obtained your answer.

**(4)**

concentration of sugar solution = ..... au

# 9. Question 4 · part c.i · 2 marks

June 2025 Paper 2

Further practice · Attempt part c.i only.

(c) (i) Explain why filter paper was wrapped around the potato cylinders at the beginning and end of the investigation.

**(2)**

.....

.....

.....

.....

(ii) The student kept the potato cylinders in the liquid for two hours.

State two other factors that should be kept constant during this investigation.

**(2)**

1 .....

.....

2 .....

.....

**(Total for Question 4 = 16 marks)**

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Working and answer

# 10. Question 2 · part a.iii · 3 marks

January 2022 Paper 1

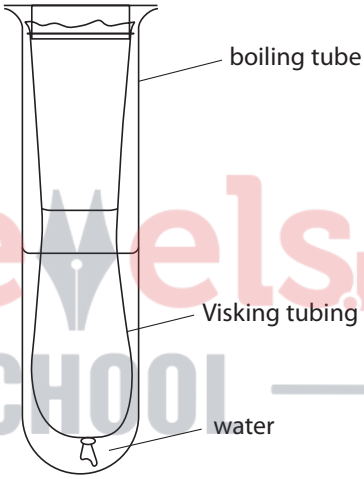
Further practice · Attempt part a.iii only.

**2** A student investigates absorption in the ileum by using a model gut made of Visking tubing.

This is the student's method.

- tie a knot in one end of a 15 cm length of Visking tubing to make a bag
- place 5 $cm^3$ of starch suspension and 5 $cm^3$ of glucose solution into the bag
- rinse the outside of the bag with water
- place the bag into a boiling tube containing distilled water
- leave the bag for one hour

The diagram shows the student's apparatus.




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After an hour the contents of the Visking tubing bag are tested for glucose and starch.

The water in the boiling tube is also tested for glucose and starch.

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## 10. Question 2 · part a.iii · 3 marks

January 2022 Paper 1

Further practice · Attempt part a.iii only.

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(a) (i) Describe how the student should test the contents of the Visking tubing bag for glucose.

**(4)**

(ii) Describe how the student should test the contents of the Visking tubing bag for starch.

**(3)**

(iii) Explain why the outside of the Visking tubing bag was rinsed with water before being placed in the boiling tube.

**(3)**

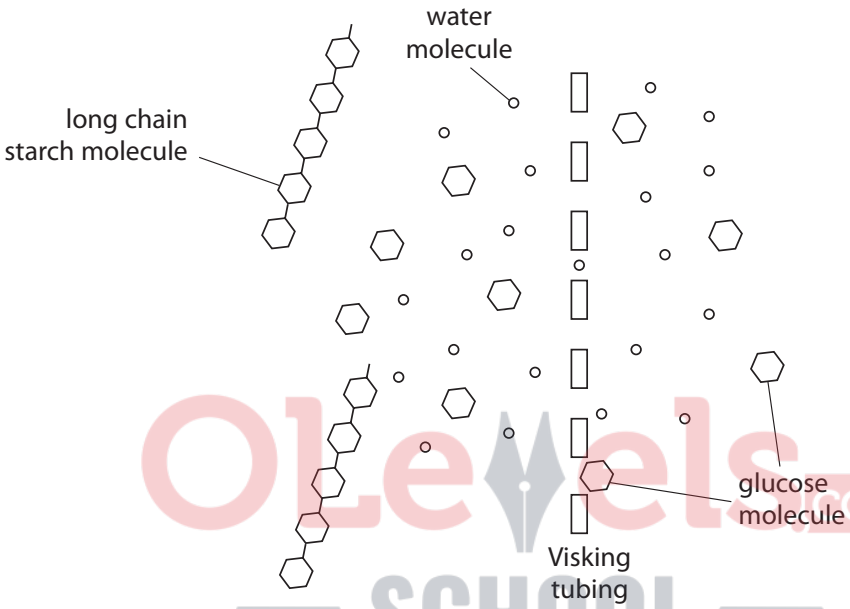
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## 10. Question 2 · part a.iii · 3 marks

January 2022 Paper 1

Further practice · Attempt part a.iii only.

(b) The student draws this diagram to explain the results obtained.




Explain the results the student obtains.

Refer to the diagram in your answer.

**(5)**

**(Total for Question 2 = 15 marks)**

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## 10. Question 2 · part a.iii · 3 marks · Working and answer

Working and answer

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# Practice workbook · Identify a measurement weakness and improve accuracy

Methods with subjective scoring, uneven bubble counts, too few or widely spaced intervals, or a contaminated reading.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-PRAC-EVALUATE-MEASUREMENT-ACCURACY

# 1. Question 1 · part a.iv · 3 marks

November 2021 Paper 1

Worked-example retry · Attempt part a.iv only.

**1** The enzyme catalase is found in the liver.

Catalase acts as a catalyst when added to hydrogen peroxide solution, causing the solution to break down into water and oxygen.

A student investigates the increase in the rate of breakdown of hydrogen peroxide into water and oxygen using catalase.

This is the student's method.

- put 100 g of liver into 100 $cm^3$ of water, and mix into a smooth paste
- put 10 $cm^3$ of the paste into a beaker at room temperature
- add one drop of hydrogen peroxide to the paste
- assess how many oxygen bubbles are produced on a scale of 0 to 5, with 5 representing most bubbles and 0 representing no bubbles

The student repeats the investigation using different conditions.

The conditions he uses are:

- liver paste heated to 70 °C
- liver paste that has been kept at 3 °C for an hour
- acidified liver paste
- liver paste with an alkali added
- liver paste with no hydrogen peroxide added

These are the student's results.

*room temperature = 5  heated to 70 °C = 0*

*kept at 3 °C = 1  acid added = 2  alkali added = 3*

*no hydrogen peroxide added = 0*

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# 1. Question 1 · part a.iv · 3 marks

November 2021 Paper 1

Worked-example retry · Attempt part a.iv only.

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(a) (i) Give the student's results in the form of a suitable table.

**(3)**

(ii) State why this investigation is not reliable.

**(1)**

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(iii) Give a reason why the results of this investigation are not accurate.

**(1)**

(iv) Describe how the investigation could be improved so that the conclusions could be more accurate and reliable.

**(3)**

# 1. Question 1 · part a.iv · 3 marks

November 2021 Paper 1

Worked-example retry · Attempt part a.iv only.

(b) (i) State the condition where the enzyme was most active.

**(1)**

(ii) State the reason for including the beaker with no hydrogen peroxide added.

**(1)**

(iii) Explain the result obtained when the conditions were acidic.

**(4)**

**(Total for Question 1 = 14 marks)**

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## 1. Question 1 · part a.iv · 3 marks · Working and answer

Working and answer

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## 2. Question 1 · part a.iii · 1 mark

November 2021 Paper 1

Further practice · Attempt part a.iii only.

**1** The enzyme catalase is found in the liver.

Catalase acts as a catalyst when added to hydrogen peroxide solution, causing the solution to break down into water and oxygen.

A student investigates the increase in the rate of breakdown of hydrogen peroxide into water and oxygen using catalase.

This is the student's method.

- put 100 g of liver into 100 $cm^3$ of water, and mix into a smooth paste
- put 10 $cm^3$ of the paste into a beaker at room temperature
- add one drop of hydrogen peroxide to the paste
- assess how many oxygen bubbles are produced on a scale of 0 to 5, with 5 representing most bubbles and 0 representing no bubbles

The student repeats the investigation using different conditions.

The conditions he uses are:

- liver paste heated to 70 °C
- liver paste that has been kept at 3 °C for an hour
- acidified liver paste
- liver paste with an alkali added
- liver paste with no hydrogen peroxide added

These are the student's results.

*room temperature = 5  heated to 70 °C = 0*

*kept at 3 °C = 1  acid added = 2  alkali added = 3*

*no hydrogen peroxide added = 0*

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## 2. Question 1 · part a.iii · 1 mark

November 2021 Paper 1

Further practice · Attempt part a.iii only.

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(a) (i) Give the student's results in the form of a suitable table.

**(3)**

(ii) State why this investigation is not reliable.

**(1)**

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(iii) Give a reason why the results of this investigation are not accurate.

**(1)**

(iv) Describe how the investigation could be improved so that the conclusions could be more accurate and reliable.

**(3)**

## 2. Question 1 · part a.iii · 1 mark · Working and answer

Working and answer

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## 3. Question 7 · part b.iv · 1 mark

November 2020 Paper 2R

Further practice · Attempt part b.iv only.

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**7** Chymosin is an enzyme found in young humans, but not in adults. It converts soluble proteins in milk into solid proteins.

(a) (i) Suggest the advantages to a young human of having chymosin in their alimentary canal.

**(2)**

(ii) Explain why chymosin is only needed in young humans.

**(2)**

(iii) Describe a test to show that the solid formed by the action of chymosin on milk is a protein.

**(3)**

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# 3. Question 7 · part b.iv · 1 mark

November 2020 Paper 2R

Further practice · Attempt part b.iv only.

(b) A student investigates the effect of carbon dioxide on the activity of chymosin.

The student bubbles different volumes of carbon dioxide gas into five samples of milk.

He then adds chymosin to each sample and records the time taken for the milk protein to become solid.

The student repeats this method three times.

The table shows the student's results.

| Number of bubbles of $CO_2$ | Time taken for chymosin to turn milk protein solid, in seconds | | | | |
|---|---|---|---|---|---|
| | Test 1 | Test 2 | Test 3 | Test 4 | Mean |
| 100 | 253 | 257 | 250 | 260 | 255 |
| 150 | 238 | 232 | 241 | 229 | |
| 200 | 216 | 214 | 219 | 211 | 215 |
| 250 | 208 | 202 | 212 | 198 | 205 |
| 300 | 210 | 200 | 199 | 311 | 203 |

(i) State three factors that the student should control.

**(3)**

1 .....

.....

2 .....

.....

3 .....

.....

(ii) Calculate the missing mean (average) time taken for 150 bubbles.

**(2)**

mean time taken = .....

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**20**

P 6 2 0 5 4 A 0 2 0 2 4

## 3. Question 7 · part b.iv · 1 mark

November 2020 Paper 2R

Further practice · Attempt part b.iv only.

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(iii) In test 4 there is an anomalous result.

State how the student deals with this result.

**(1)**

.....

.....

(iv) State why recording the number of bubbles may produce inaccurate results.

**(1)**

.....

.....

**(Total for Question 7 = 14 marks)**

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Working and answer

# 4. Question 7 · part b.ii · 2 marks

June 2019 Paper 1R

Further practice · Attempt part b.ii only.

**7** Cheese is processed from milk. During this process, an enzyme is used to convert the lactose in milk into lactic acid.

A student uses this method to investigate the process.

- dissolve lactose in water to produce a solution
- put the lactose solution into three different test tubes
- keep the first test tube at 2 °C, the second test tube at 25 °C and the third test tube at 55 °C
- add enzyme solution to the lactose solution in each test tube
- measure the amount of lactic acid in the solution every 10 hours for a total of 60 hours

The table shows the student's results.

| Time after adding enzyme in hours | Amount of lactic acid in the solution in arbitrary units | | |
|---|---|---|---|
| | 2 °C | 25 °C | 55 °C |
| 0 | 0.1 | 0.1 | 0.1 |
| 10 | 0.1 | 0.2 | 0.2 |
| 20 | 0.1 | 0.3 | 0.2 |
| 30 | 0.2 | 0.8 | 0.4 |
| 40 | 0.3 | 1.8 | 0.7 |
| 50 | 0.4 | 2.2 | 1.0 |
| 60 | 0.6 | 2.3 | 1.3 |

(a) Describe what additional information another student would need to repeat this investigation.

**(4)**

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**20**

# 4. Question 7 · part b.ii · 2 marks

June 2019 Paper 1R

Further practice · Attempt part b.ii only.

(b) (i) Estimate the optimum temperature for this enzyme.

**(1)**

(ii) Explain why it is not possible to give an accurate value for the optimum temperature using the student's method.

**(2)**

(c) The student continued the investigation for a further 60 hours.

Explain the results that would be obtained.

**(3)**

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## 4. Question 7 · part b.ii · 2 marks · Working and answer

Working and answer

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# 5. Question 6 · part a.iii · 2 marks

January 2023 Paper 1

Further practice · Attempt part a.iii only.

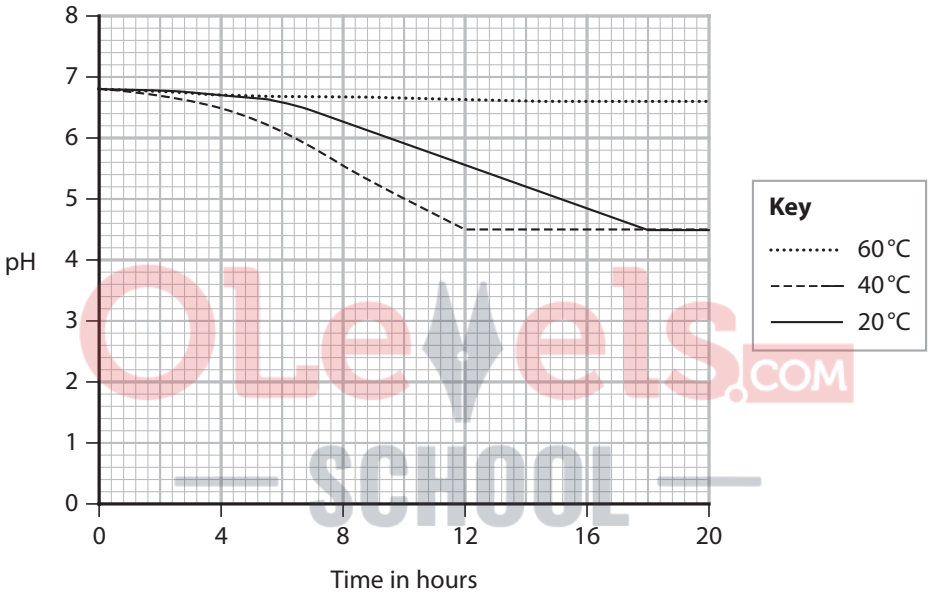
**6** Yoghurt can be produced by incubating milk with a small amount of live yoghurt.

The live yoghurt contains *Lactobacillus* bacteria.

Enzymes convert lactose in the milk into lactic acid causing the pH to decrease to around pH 4.4.

In an investigation, the change in pH as yoghurt is produced was measured during 20 hours at three different temperatures.

The graph shows the results.




(a) (i) Calculate the difference in the time taken to produce yoghurt at 40 °C instead of 20 °C.

**(2)**

time = ..... hours

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## 5. Question 6 · part a.iii · 2 marks

January 2023 Paper 1

Further practice · Attempt part a.iii only.

(ii) Give two variables that need to be controlled in this investigation.

**(2)**

1 .....

.....

2 .....

.....

(iii) Describe how this investigation could be improved to find the optimum temperature for the production of yoghurt.

**(2)**

.....

.....

.....

.....

.....

(iv) In some methods to produce yoghurt, the milk is boiled and then cooled before the bacteria are added.

Give the reason for boiling and the reason for cooling the milk.

**(2)**

boiling

.....

.....

cooling

.....

.....

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## 5. Question 6 · part a.iii · 2 marks · Working and answer

Working and answer

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# 6. Question 3 · part c.ii · 1 mark

June 2019 Paper 2

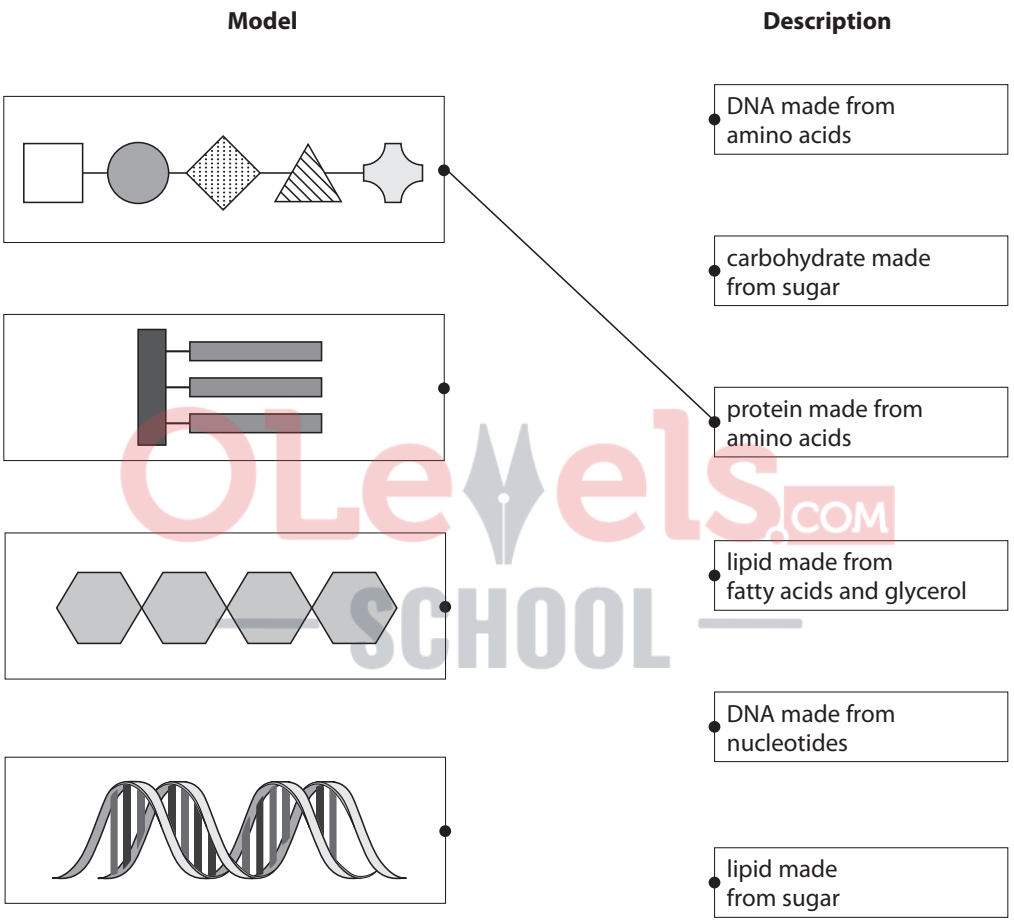
Further practice · Attempt part c.ii only.

**3** (a) The diagram shows models of four biological molecules.

Draw a straight line from each model to its correct description.

One has been done for you.

**(3)**




(b) Which body organ produces enzymes that break down protein?

**(1)**

- ☒ **A** gall bladder
- ☒ **B** large intestine
- ☒ **C** mouth
- ☒ **D** stomach

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# 6. Question 3 · part c.ii · 1 mark

June 2019 Paper 2

Further practice · Attempt part c.ii only.

(c) A student uses this method to test different substances for protein.

- grind solid substances into small pieces
- place substances into separate test tubes
- add a few drops of Biuret reagent to each substance
- record the colour change for each substance

The table shows the student's results.

| Substance | Colour after Biuret test |
|---|---|
| milk | purple |
| pasta | blue |
| lemon juice | blue |
| cheese | purple |
| distilled water | purple |

(i) In the student's test, how many of the substances give a positive result for protein?

**(1)**

.....

(ii) The result for distilled water is incorrect.

Suggest one mistake the student could have made to get this incorrect result.

**(1)**

.....

.....

(d) The table shows an incomplete risk assessment for the Biuret test.

Complete the table by describing how to reduce the risk of each hazard.

**(2)**

| Hazard | Reducing risk |
|---|---|
| stools – trip hazard | keep stools under bench |
| broken glass – cuts | |
| Biuret reagent – irritant | |

**(Total for Question 3 = 8 marks)**

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## 6. Question 3 · part c.ii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

# Practice workbook · Plan or specify a single-factor laboratory method

Bench investigations of enzyme temperature, osmosis in potato, diffusion into agar cubes, and the detail missing from a written method.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-PRAC-PLAN-CONTROLLED-LAB-METHOD

# 1. Question 5 · part a.i · 5 marks

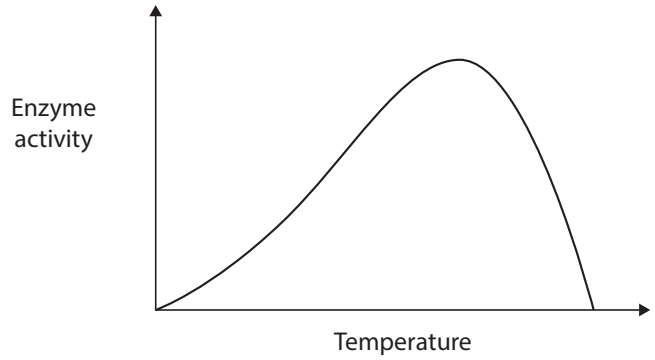
June 2024 Paper 1

Worked-example retry · Attempt part a.i only.

**5** (a) A group of students investigate the effect of temperature on enzyme activity.

The students use the results they obtain to plot a graph.

The graph shows the students' results.




(Source: adapted from https://ib.bioninja.com.au/standard-level/topic-2-molecular-biology/25-enzymes/enzyme-activity.html)

(i) Design a method the students could use to obtain the results shown in the graph.

**(5)**

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# 1. Question 5 · part a.i · 5 marks

June 2024 Paper 1

Worked-example retry · Attempt part a.i only.

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(ii) State the independent variable in this investigation.

**(1)**

.....

(b) Explain how enzyme activity is affected by low temperatures.

**(4)**

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**(Total for Question 5 = 10 marks)**

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## 1. Question 5 · part a.i · 5 marks · Working and answer

Working and answer

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## 2. Question 5 · part b · 5 marks

June 2019 Paper 2R

Further practice · Attempt part b only.

**5** Substances move in and out of cells by diffusion, osmosis and active transport.

(a) Describe the process of active transport.

**(2)**

(b) Water moves in and out of potato cells by osmosis.

Describe a method to determine the concentration of the contents of potato cells.

A large potato is available and also sucrose solutions of various concentrations.

**(5)**

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## 2. Question 5 · part b · 5 marks

June 2019 Paper 2R

Further practice · Attempt part b only.

(c) The table shows results obtained from an investigation to determine the concentration of the contents of potato cells.

| | Concentration of sucrose solution in mol per dm³ | | | | |
|---|---|---|---|---|---|
| | **0.2** | **0.4** | **0.6** | **0.8** | **1.0** |
| Initial mass in g | 8.4 | 7.9 | 8.7 | 8.3 | 8.4 |
| Mass after soaking in g | 8.7 | 8.0 | 8.2 | 7.5 | 7.3 |
| Change in mass in g | + 0.3 | + 0.1 | – 0.5 | – 0.8 | – 1.1 |
| Percentage change in mass | + 3.6 | | – 5.8 | – 9.6 | – 13.1 |

(i) Calculate the percentage change in mass for 0.4 mol per dm³.

**(2)**

.....%

(ii) Explain why it is necessary to calculate the percentage change in mass rather than using the change in mass in g.

**(2)**

.....

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P 6 0 1 8 7 A 0 1 4 2 4

## 2. Question 5 · part b · 5 marks · Working and answer

Working and answer

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# 3. Question 3 · part b.i · 2 marks

January 2023 Paper 2

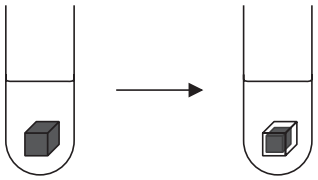
Further practice · Attempt part b.i only.

**3** A student investigates diffusion using agar cubes and an acid.

The agar cube contains an alkali and a pH indicator.

The indicator is pink in alkali and colourless in acid.

Acid diffuses into the cube, causing the cube to change from pink to colourless from the edge.



agar cube when first placed in acid

agar cube starts to become colourless

(a) The cube the student placed in the acid had a side length of 2 mm.

(i) Calculate the surface area to volume ratio of the cube.

Your answer should be in the form n : 1

**(3)**

surface area to volume ratio = .....

(ii) The cube took 1 minute and 32 seconds to become colourless.

Which of these shows the time in seconds?

**(1)**

- ☒ **A** 62
- ☒ **B** 82
- ☒ **C** 92
- ☒ **D** 102

(iii) State what is meant by the term **diffusion**.

**(1)**

.....

.....

.....

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P 7 2 6 0 2 A 0 8 2 8

# 3. Question 3 · part b.i · 2 marks

January 2023 Paper 2

Further practice · Attempt part b.i only.

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(iv) Explain the effect of increasing the surface area to volume ratio on the time taken for the cube to become colourless.

**(2)**

(b) A student wants to investigate the effect of temperature on the rate of diffusion into the agar cube.

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(i) Describe a method for this investigation using the apparatus shown in the diagram.

**(2)**

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(ii) Give two variables the student would need to control in their investigation.

**(2)**

1 .....

2 .....

**(Total for Question 3 = 11 marks)**

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## 3. Question 3 · part b.i · 2 marks · Working and answer

Working and answer

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# 4. Question 7 · part a · 4 marks

June 2019 Paper 1R

Further practice · Attempt part a only.

**7** Cheese is processed from milk. During this process, an enzyme is used to convert the lactose in milk into lactic acid.

A student uses this method to investigate the process.

- dissolve lactose in water to produce a solution
- put the lactose solution into three different test tubes
- keep the first test tube at 2 °C, the second test tube at 25 °C and the third test tube at 55 °C
- add enzyme solution to the lactose solution in each test tube
- measure the amount of lactic acid in the solution every 10 hours for a total of 60 hours

The table shows the student's results.

| Time after adding enzyme in hours | Amount of lactic acid in the solution in arbitrary units | | |
|---|---|---|---|
| | 2 °C | 25 °C | 55 °C |
| 0 | 0.1 | 0.1 | 0.1 |
| 10 | 0.1 | 0.2 | 0.2 |
| 20 | 0.1 | 0.3 | 0.2 |
| 30 | 0.2 | 0.8 | 0.4 |
| 40 | 0.3 | 1.8 | 0.7 |
| 50 | 0.4 | 2.2 | 1.0 |
| 60 | 0.6 | 2.3 | 1.3 |

(a) Describe what additional information another student would need to repeat this investigation.

**(4)**

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# 4. Question 7 · part a · 4 marks

June 2019 Paper 1R

Further practice · Attempt part a only.

(b) (i) Estimate the optimum temperature for this enzyme.

**(1)**

(ii) Explain why it is not possible to give an accurate value for the optimum temperature using the student's method.

**(2)**

(c) The student continued the investigation for a further 60 hours.

Explain the results that would be obtained.

**(3)**

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## 4. Question 7 · part a · 4 marks · Working and answer

Working and answer

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